# H2 Math Summary

Example Questions

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# 1 Systems of Linear Equations

**1.1** Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

| Café | Discounts given for each package | | | Total price after the discount |
|---|---|---|---|---|
| | Blend A | Blend B | Blend C | |
| **M Café** | 15% | 10% | 5% | \$161.80 |
| **I Café** | 7% | 5% | 2% | \$169.56 |

**(i)** Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

**[MI 2025 Prelim P1 Q4]**

| Solution | | Comments |
|---|---|---|
| **(i)**<br>**[3]** | Let the usual prices be \$$a$, \$$b$, \$$c$.<br>$a+b+c=176 \quad ---(1)$<br>$0.85a+0.90b+0.95c=161.80 \quad ---(2)$<br>$0.93a+0.95b+0.98c=169.56 \quad ---(3)$<br>Using GC, $a=32,\ b=44,\ c=100$<br>$\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100 | |

**1.2** A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where $a$, $b$, $c$ and $d$ are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of $a$, $b$, $c$ and $d$. [4]

**[SAJC 2025 Prelim P1 Q1]**

| **Solution** | | **Comments** |
|---|---|---|
| **[4]** | $\mathrm{f}'(x)=3ax^2+2bx+c$<br>$\mathrm{f}'(-1)=0$<br>$3a-2b+c=0 \quad ---(1)$<br>$\mathrm{f}(2)=10$<br>$8a+4b+2c+d=10 \quad ---(2)$<br>$\mathrm{f}(-2)=-2$<br>$-8a+4b-2c+d=-2 \quad ---(3)$<br>$\int_{-2}^{2} (ax^3+bx^2+cx+d)\,\mathrm{d}x=16$<br>$\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$<br>$(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$<br>$\frac{16}{3}b+4d=16 \quad ---(4)$<br>Using GC, $a=3,\ b=0,\ c=-9,\ d=4$ | |

Two linear equations are independent if and only if neither equation is a linear combination of the other.

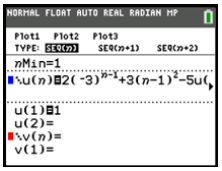
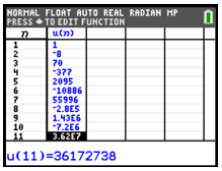
**1.3** A sequence of real numbers, $x_n$, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of $a$, $b$ and $c$. [3]

Hence, find the values of $x_{10}$ and $x_{11}$. [2]

**[HCI 2025 Prelim P1 Q2]**

| **Solution** | | **Comments** |
|---|---|---|
| **[2]** | $n = 1$<br>$x_2 = a(-3)^1 + b(1)^2 + cx_1$<br>$-8 = -3a + b + c \quad ---(1)$<br>$n = 2$<br>$x_3 = a(-3)^2 + b(2)^2 + cx_2$<br>$70 = 9a + 4b - 8c \quad ---(2)$<br>$n = 3$<br>$x_4 = a(-3)^3 + b(3)^2 + cx_3$<br>$-377 = -27a + 9b + 70c \quad ---(3)$<br>Using GC, $a = 2,\ b = 3,\ c = -5$<br>$x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$<br>Using GC,<br>$x_{10} = -7\,210\,868$<br>$x_{11} = 36\,172\,738$<br>  <br><br>  <br> | |

**1.4** The lines $l_1$ and $l_2$ have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where $\lambda$ and $\mu$ are parameters.

**(i)** Show that $l_1$ and $l_2$ are skew lines. [2]

**[SAJC 2025 Prelim P1 Q3]**

| **Solution** | | **Comments** |
|---|---|---|
| **(i)** **[2]** | $1-2\lambda = 2\mu$ $---(1)$<br>$2+\lambda = 5-2\mu$ $---(2)$<br>$2\lambda = -8+3\mu$ $---(3)$<br>Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$<br>Sub into (3):<br>$\text{LHS} = 2(-2) = -4$<br>$\text{RHS} = -8+3\left(\frac{5}{2}\right) = -\frac{1}{2}$<br>$\text{LHS} \neq \text{RHS},$ no solution<br>$\therefore l_1$ and $l_2$ do not intersect<br>$\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$<br>$\therefore l_1$ and $l_2$ are not parallel<br>$\therefore l_1$ and $l_2$ are skew (shown) | |

**1.5** A movie theatre supports a charity event by providing discounted movie tickets at the following prices based on age group.

| Age group | Discounted ticket price |
|---|---|
| $\leqslant 12$ years | \$6 |
| 13 to 49 years | \$9 |
| $\geqslant 50$ years (senior citizen) | \$4 |

The event organiser plans to spend \$700 to sponsor 100 participants of which at least 35 of them are senior citizens. Find the possible number of participants for each age group. [4]

**[TJC 2022 Prelim P2 Q1]**

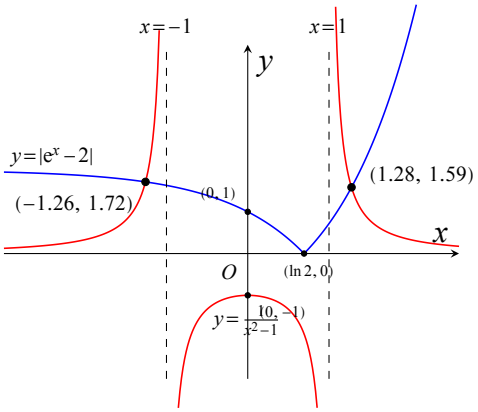
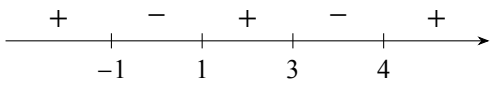
| **Solution** | | **Comments** |
|---|---|---|
| **[4]** | Let $x$, $y$, $z$ be the number of participants aged $\leqslant 12$, 13 to 49 and $\geqslant 50$ years respectively.<br>$\left.\begin{aligned} x+y+z=100 \quad &---(1) \\ 6x+9y+4z=700 \quad &---(2) \end{aligned}\right\} \Rightarrow$ Using GC, $x=\frac{200}{3}-\frac{5}{3}z,\ y=\frac{100}{3}+\frac{2}{3}z$<br>$z \geqslant 35$<br>$\because x \geqslant 0$<br>$\frac{200}{3}-\frac{5}{3}z \geqslant 0$<br>$z \leqslant 40$<br>$\therefore 35 \leqslant z \leqslant 40$<br>$x$, $y$, $z$ are non-negative integers. Using GC table:<br>$z=37 \Rightarrow x=5,\ y=58$<br>$z=40 \Rightarrow x=0,\ y=60$<br>$\therefore (x, y, z)=(5, 58, 37)$ or $(0, 60, 40)$ | |

# 2 Inequalities

**2.1** **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

**(b)** Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

**[RVHS 2026 Prelim P1 Q2]**

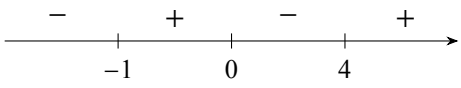
| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** |  <br>Using GC, intersections at $x=-1.2581$, $x=1.2770$<br>$\therefore x<-1.26$ or $-1<x<1$ or $x>1.28$ | |
| **(b)** **[3]** | $\dfrac{x+2}{x-1} - \dfrac{2}{x-3} \geqslant 0$<br>$\dfrac{(x+2)(x-3)-2(x-1)}{(x-1)(x-3)} \geqslant 0$<br>$\dfrac{x^2-3x-4}{(x-1)(x-3)} \geqslant 0$<br>$\dfrac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$<br>  <br>$\therefore x \leqslant -1$ or $1<x<3$ or $x \geqslant 4$ | |

**2.2** Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

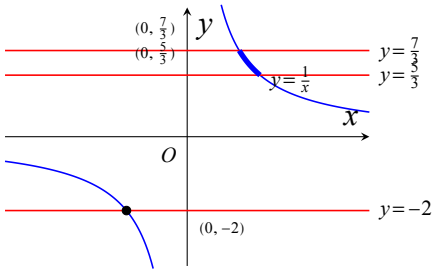
**[ACJC 2026 Prelim P1 Q1]**

| Solution | | Comments |
|---|---|---|
| **[4]** | $x \geqslant 3x-6$<br>$x \leqslant 3 \quad ---(1)$<br>$x^2+2x+5=(x+1)^2+4>0$ for all real $x$<br>$\frac{x(5x+9)}{x^2+2x+5} > x$<br>$5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$<br>$x^3-3x^2-4x<0$<br>$x(x-4)(x+1)<0$<br>  <br><br>$x<-1$ or $0<x<4 \quad ---(2)$<br>$(1)\cap(2)$ :<br>$\therefore x<-1$ or $0<x\leqslant 3$ | |

**2.3** **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

**(ii)** Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

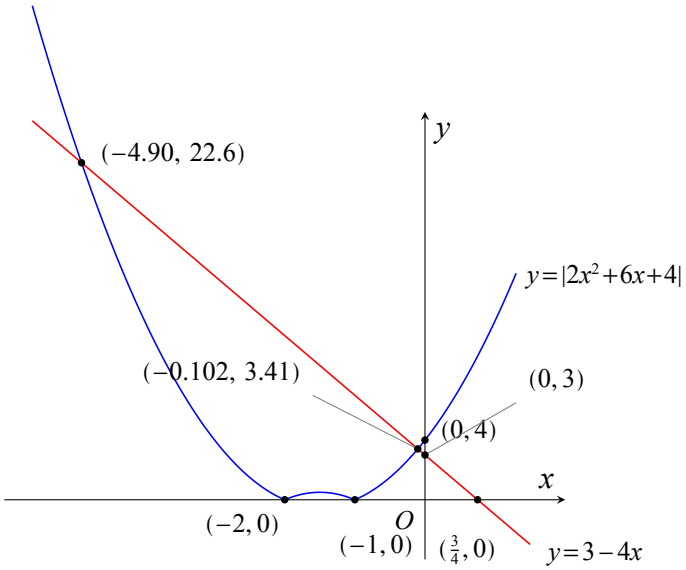
**[SAJC 2025 Prelim P2 Q1]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[4]** | $3\|x-2\|\|x+2\| \leqslant \|x+2\|$<br>$x=-2:\quad 0 \leqslant 0$ (true)<br>$x \neq -2:\ \|x+2\| > 0$<br>$3\|x-2\| \leqslant 1$<br>$-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$<br>$\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$<br>$\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ | |
| **(ii)** **[2]** | $\dfrac{3}{x^2}\,\|1-4x^2\| = 3\left\|\dfrac{1}{x^2}-4\right\| \quad (\because x^2>0)$<br>$\left\|\dfrac{1+2x}{x}\right\| = \left\|\dfrac{1}{x}+2\right\|$<br>$3\left\|\left(\dfrac{1}{x}\right)^2-4\right\| \leqslant \left\|\dfrac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$<br>$\dfrac{1}{x}=-2$ or $\dfrac{5}{3} \leqslant \dfrac{1}{x} \leqslant \dfrac{7}{3}$<br>  <br><br>$\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$ | |

**2.4** **(i)** On the same axes, sketch the curves with equations $y=|2x^2+6x+4|$ and $y=3-4x$, indicating any intercepts with the axes and points of intersection. Hence solve the inequality $3-4x<|2x^2+6x+4|$. [4]

**(ii)** Find the exact area bounded by the graphs of $y=3-4x$, $y=|2x^2+6x+4|$, $x=-3$ and $x=-1$. [3]

**[SAJC 2019 Prelim P1 Q2]**

| **Solution** | | **Comments** |
|---|---|---|
| **(i)** **[4]** |  | |
| | $3-4x=2x^2+6x+4$<br>$2x^2+10x+1=0$<br>$x=\dfrac{-5\pm\sqrt{23}}{2}$<br>$\therefore x<\dfrac{-5-\sqrt{23}}{2}$ or $x>\dfrac{-5+\sqrt{23}}{2}$ | |
| **(ii)** **[3]** | $\text{Area}=\int_{-3}^{-1}(3-4x-\lvert2x^2+6x+4\rvert)\,dx$<br>$=\int_{-3}^{-2}(3-4x-2x^2-6x-4)\,dx+\int_{-2}^{-1}(3-4x+2x^2+6x+4)\,dx$<br>$=\int_{-3}^{-2}(-2x^2-10x-1)\,dx+\int_{-2}^{-1}(2x^2+2x+7)\,dx$<br>$=\left[-\frac{2}{3}x^3-5x^2-x\right]_{-3}^{-2}+\left[\frac{2}{3}x^3+x^2+7x\right]_{-2}^{-1}$<br>$=\left(\frac{16}{3}-20+2\right)-(18-45+3)+\left(-\frac{2}{3}+1-7\right)-\left(-\frac{16}{3}+4-14\right)$<br>$=20$ | |

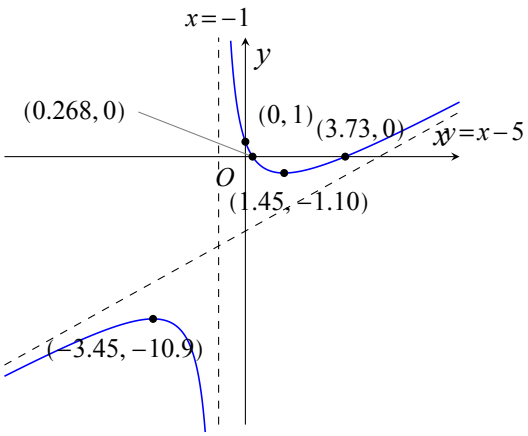
# 3 Graphing Techniques

**3.1** A curve $C$ has equation $y=\dfrac{x^2-4x+1}{x+1}$, $x\neq -1$.

**(a)** Prove, using an algebraic method, that $y=\dfrac{x^2-4x+1}{x+1}$ cannot lie between two certain values, to be determined in exact form. [3]

**(b)** Sketch $C$, stating the equations of any asymptotes, and the coordinates of any turning points and any axial intercepts. [4]

**[VJC 2025 Promo Q9(a), (b)]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[3]** | $y=\dfrac{x^2-4x+1}{x+1}$<br>$yx+y=x^2-4x+1$<br>$x^2+(-y-4)x+(1-y)=0$<br>No real solution for $x$: $(-y-4)^2-4(1-y)<0$<br>$y^2+8y+16-4+4y<0$<br>$y^2+12y+12<0$<br>$(y+6)^2-24<0$<br>$\left(y+6-\sqrt{24}\right)\left(y+6+\sqrt{24}\right)<0$<br>$-6-2\sqrt{6}<y<-6+2\sqrt{6}$<br>$\therefore y$ cannot lie between $-6-2\sqrt{6}$ and $-6+2\sqrt{6}$ | |
| **(b)**<br>**[4]** | $y=x-5+\dfrac{6}{x+1}$<br>Asymptotes: $x=-1,\ y=x-5$<br>$x=0$: $y=1$<br>$y=0$: $x^2-4x+1=0 \Rightarrow x=0.268,\ 3.73$<br>Turning points: $(1.45,-1.10),\ (-3.45,-10.9)$<br>  <br> | |

**3.2** A curve $C$ has equation $y=ax+b+\dfrac{2b-a}{x+3}$, where $a$ and $b$ are constants such that $a>0$, $b\neq\dfrac{1}{2}a$ and $x\neq -3$.

**(a)** If $C$ has no stationary points, use differentiation to find the relationship between $a$ and $b$. [3]

**(b)** It is now given that $b=a$.

**(i)** Find the $x$-coordinates of the turning points. [2]

**(ii)** Hence sketch $C$, stating the equations of any asymptotes and the coordinates of the axial intercepts and turning points. [3]

**(iii)** Verify that the point of intersection of the two asymptotes of $C$ lies on the line $y=kx+3k-2a$.

Hence, using the graph in part (b)(ii), find the range of values of $k$ in terms of $a$, such that the equation $x(k-a)+3k-2a-b-\dfrac{2b-a}{x+3}=0$ has real roots. [2]

**[SAJC 2025 Promo Q7]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $\dfrac{dy}{dx}=a-\dfrac{2b-a}{(x+3)^2}$<br>$\dfrac{dy}{dx}=0:\quad a-\dfrac{2b-a}{(x+3)^2}=0$<br>$(x+3)^2=\dfrac{2b-a}{a}\qquad ---(1)$<br>No stationary points $\Rightarrow\ \dfrac{2b-a}{a}<0$<br>Since $a>0,\quad 2b-a<0$<br>$\therefore a>2b$ | |
| **(b)(i)** **[2]** | $b=a:\quad (x+3)^2=\dfrac{2a-a}{a}=1$<br>$x+3=1$ or $-1$<br>$x=-2$ or $-4$ | |

**(b)(ii)**
**[3]**

$$y = ax + a + \frac{a}{x+3}$$

Asymptotes: $x = -3,\ y = ax + a$

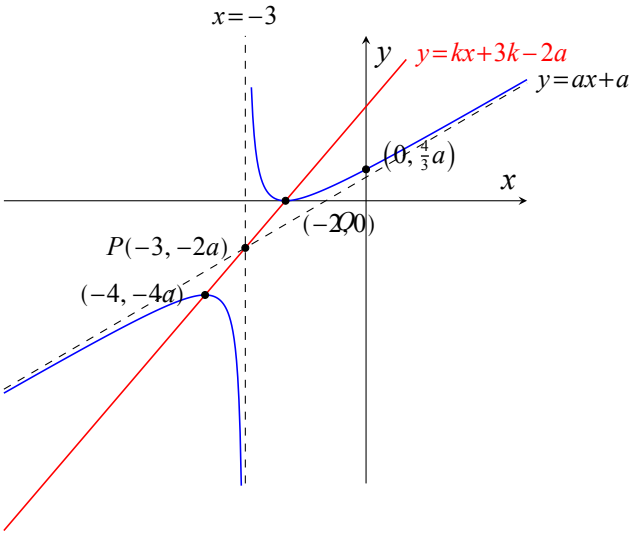
$x = -2 : y = 0; \quad x = -4 : y = -4a$

Turning points: $(-2, 0),\ (-4, -4a)$

$$x = 0 : y = \frac{4}{3}a$$

$$y = 0 : ax + a + \frac{a}{x+3} = 0 \Rightarrow (x+1)(x+3) + 1 = 0$$

$$x^2 + 4x + 4 = 0 \Rightarrow (x+2)^2 = 0 \Rightarrow x = -2$$




**(b)(iii)**
**[2]**

$x = -3$ and $y = ax + a$: $\quad y = -3a + a = -2a$

Intersection of asymptotes: $P(-3, -2a)$

$x = -3 : \quad \text{RHS} = k(-3) + 3k - 2a = -2a = \text{LHS}$

$\therefore P$ lies on the line $y = kx + 3k - 2a$

$$x(k-a) + 3k - 2a - b - \frac{2b-a}{x+3} = 0$$

$$kx + 3k - 2a = ax + b + \frac{2b-a}{x+3}$$

Line through $P$ meets $C \Rightarrow k > a$

**3.3** A curve $C$ has equation $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}$, where $x>-2$ and $k$ is a non-zero real constant.
It is given that the function f is strictly increasing.

**(a)** Find the range of values of $k$. [3]

**(b)** Sketch the curve $C$, giving the coordinates of any points where $C$ crosses the $x$- and $y$-axes and the equations of any asymptotes. [3]

**(c)** By adding a suitable curve to the graph of the curve $C$ in (b), deduce the number of distinct real root(s) of the equation $(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36)$. [3]

**[DHS 2025 Promo Q8]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}=3x+\dfrac{k}{x+2}$ | |
| | $\mathrm{f}'(x)=3-\dfrac{k}{(x+2)^2}=\dfrac{3x^2+12x+12-k}{(x+2)^2}$ | |
| | f strictly increasing $\Rightarrow$ $\mathrm{f}'(x)>0$ | |
| | $3x^2+12x+12-k>0 \quad \because (x+2)^2>0$ for $x>-2$ | |
| | $3(x+2)^2-k>0$ | |
| | Since $3(x+2)^2-k>-k$ for all $x>-2$, $\quad -k\geqslant 0$ | |
| | $k\neq 0 \Rightarrow k<0$ | |
| **(b)** **[3]** | $y=3x+\dfrac{k}{x+2}$ | |
| | Asymptotes: $y=3x,\ x=-2$ | |
| | $x=0:\ y=\dfrac{k}{2}$ | |
| | $y=0:\ 3x^2+6x+k=0 \Rightarrow x=\dfrac{-6\pm\sqrt{36-12k}}{6}=-1\pm\dfrac{\sqrt{9-3k}}{3}$ | |
| | $x>-2 \Rightarrow x=-1+\dfrac{\sqrt{9-3k}}{3}$ | |

**(c)**
**[3]**

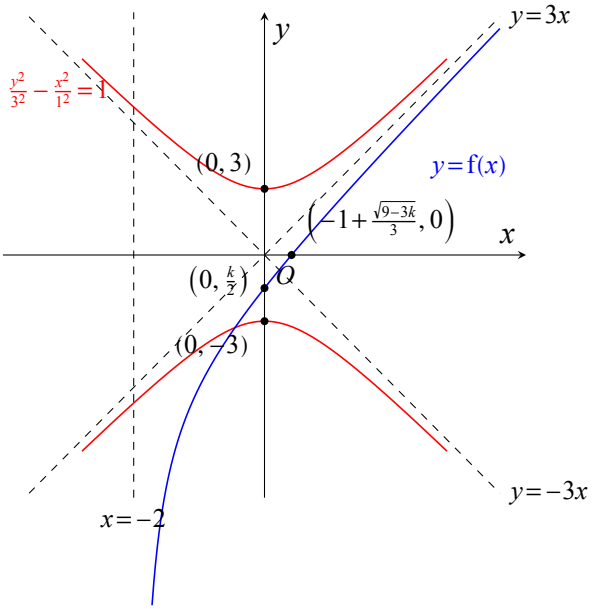
$$(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36) \quad ---(*)$$

$$(3x^2+6x+k)^2=9\,(x^2+1)\,(x+2)^2$$

$$\left(\frac{3x^2+6x+k}{x+2}\right)^2-9x^2=9$$

Draw the hyperbola $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$

Asymptotes $y=\pm 3x$, vertices $(0,\pm 3)$




$y=\mathrm{f}(x)$ and $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$ have one intersection point

$\therefore$ $(*)$ has one real root

**3.4** The curve $C$ is defined by the parametric equations $x=6t-5$ and $y=2t^2+1$, where $t \geqslant \frac{5}{6}$.

**(a)** Sketch the curve $C$, labelling the exact coordinates of any axial intercept(s). [1]

**(b)** Find the equation of the normal to $C$ at the point $P$ where $t=1$. [2]

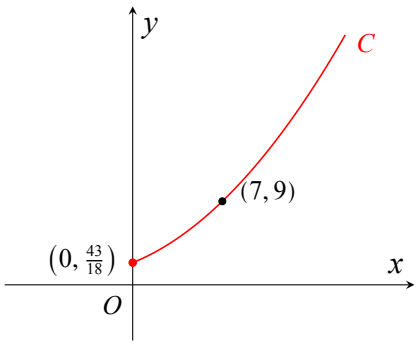
**(c)** Find the acute angle between the normal to $C$ at the point $P$ where $t=1$ and the tangent to $C$ at the point $M$ where $t=3$. [2]

**(d)** The curve $C$ is translated 5 units in the positive $x$-direction and translated 3 units in the negative $y$-direction, to form the curve $D$. Find the equation of $D$ in parametric form. [2]

The curve $E$ is defined by the parametric equations $x=7u$ and $y=\frac{9}{u}$, where $u \neq 0$.

**(e)** Show that at the point of intersection of the curves $C$ and $E$, $6t^3-5t^2+3t-34=0$. Deduce that there is only one point of intersection and find the coordinates of this point. [4]

**[RVHS 2025 Prelim P1 Q12]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)**<br>**[1]** | $t=\frac{5}{6}$: $x=6\left(\frac{5}{6}\right)-5=0$, $y=2\left(\frac{5}{6}\right)^2+1=\frac{43}{18}$<br>Endpoint ($y$-intercept): $\left(0, \frac{43}{18}\right)$<br>  <br> | |

**(b)** **[2]**

$$\frac{dx}{dt}=6, \quad \frac{dy}{dt}=4t$$

$$\frac{dy}{dx}=\frac{4t}{6}=\frac{2}{3}t$$

$t=1$: $\quad \frac{dy}{dx}=\frac{2}{3} \Rightarrow$ gradient of normal $=-\frac{3}{2}$

$t=1$: $\quad x=1,\ y=3$

$$y-3=-\frac{3}{2}(x-1)$$

$$y=-\frac{3}{2}x+\frac{9}{2}$$

**(c)** **[2]**

$t=3$: $\quad$ gradient of tangent $=\frac{2}{3}(3)=2$

$$\theta_1+\theta_2=\tan^{-1}(2)-\tan^{-1}\left(-\frac{3}{2}\right)=2.089942 \text{ or } 119.7^\circ$$

$$\text{Acute angle}=\pi-2.089942 \text{ or } 180^\circ-119.7^\circ$$

$$=1.05 \text{ or } 60.3^\circ$$

**(d)** **[2]**

$x-5=6t-5 \Rightarrow x=6t$

$y+3=2t^2+1 \Rightarrow y=2t^2-2$

$D$: $\quad x=6t,\ y=2t^2-2$

**(e)** **[4]**

$$6t-5=7u \Rightarrow u=\frac{6t-5}{7} \quad ---(1)$$

$$2t^2+1=\frac{9}{u} \quad ---(2)$$

Sub (1) into (2): $2t^2+1=\frac{63}{6t-5}$

$$(2t^2+1)(6t-5)=63$$

$$12t^3-10t^2+6t-68=0$$

$$6t^3-5t^2+3t-34=0 \quad \text{(shown)}$$

$$(t-2)(6t^2+7t+17)=0$$

$6t^2+7t+17$: $b^2-4ac=7^2-4(6)(17)=-359<0 \Rightarrow$ no real roots

$\therefore t=2$ only $\Rightarrow$ only one point of intersection

$t=2$: $\quad x=6(2)-5=7,\ y=2(2)^2+1=9$

Point of intersection: $(7, 9)$

**3.5** The parametric equations of the curve $C$ are

$$x=\tan\theta-1 \quad \text{and} \quad y=2\sec\theta, \quad \text{where } -\frac{\pi}{2}<\theta<\frac{\pi}{2}.$$

**(a)** Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of $\theta$, leaving your answer in a single trigonometric function. [2]

**(b)** Find the equation of the tangent to $C$ at the point where it cuts the $y$-axis, leaving your answer in exact form. [3]

**(c)** Find the Cartesian equation of $C$. [1]

**(d)** Find the equations of the asymptotes and state the coordinates of any turning point(s). [3]

**(e)** The region bounded by the tangent found in part (b), the curve $C$ and the line $x=-1$ is rotated completely about the $x$-axis. Calculate the volume obtained. [3]

**[HCI 2024 Promo Q12]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[2]** | $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}\theta}\div\frac{\mathrm{d}x}{\mathrm{d}\theta}=\frac{2\sec\theta\tan\theta}{\sec^2\theta}=2\sin\theta$ | |
| **(b)** **[3]** | $x=0$: $\tan\theta-1=0 \Rightarrow \tan\theta=1$<br>$\theta=\frac{\pi}{4}$ since $-\frac{\pi}{2}<\theta<\frac{\pi}{2}$<br>$\theta=\frac{\pi}{4}$: $y=\frac{2}{\cos\frac{\pi}{4}}=2\sqrt{2}$, $\frac{\mathrm{d}y}{\mathrm{d}x}=2\sin\frac{\pi}{4}=\sqrt{2}$<br>Tangent: $y=\sqrt{2}x+2\sqrt{2}$ | |
| **(c)** **[1]** | $\tan^2\theta+1=\sec^2\theta$<br>$(x+1)^2+1=\left(\frac{y}{2}\right)^2$<br>$-\frac{\pi}{2}<\theta<\frac{\pi}{2} \Rightarrow 0<\cos\theta\leqslant 1 \Rightarrow \sec\theta\geqslant 1$<br>$\therefore (x+1)^2+1=\left(\frac{y}{2}\right)^2, \quad y\geqslant 2$ | |

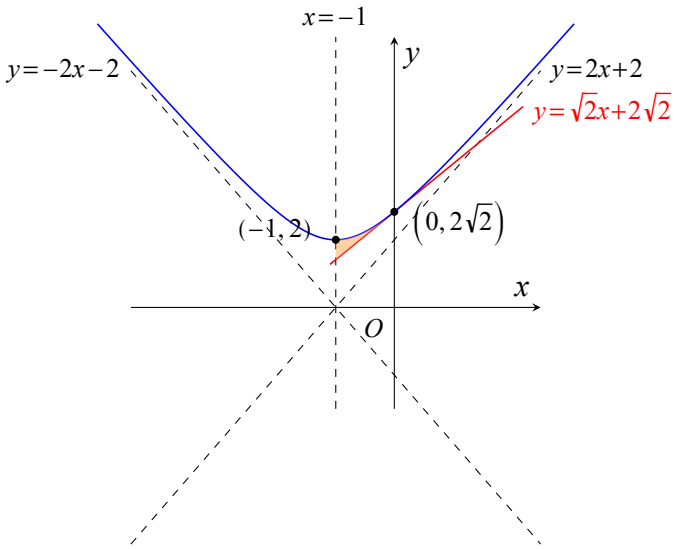
**(d)**
**[3]**

Asymptotes: $\left(\frac{y}{2}\right)^2 = (x+1)^2$

$$\frac{y}{2} = \pm(x+1)$$

$y = 2x+2$ or $y = -2x-2$

Centre $(-1, 0)$, $y \geqslant 2 \Rightarrow$ turning point $(-1, 2)$




**(e)**
**[3]**

$$\frac{y^2}{4} = (x+1)^2 + 1$$

$$y^2 = 4(x+1)^2 + 4$$

$$\text{Volume} = \pi \int_{-1}^{0} \left[ 4\left((x+1)^2+1\right) - \left(\sqrt{2}x + 2\sqrt{2}\right)^2 \right] dx$$

$$= 2.09 \text{ units}^3$$

# 4 Graphing Transformations

**4.1** A curve with equation $y=\mathrm{h}(x)$ undergoes the following sequence of transformations.

**A**: A translation of 2 units in the positive direction of the $x$-axis.

**B**: A scaling parallel to the $x$-axis by factor of 3.

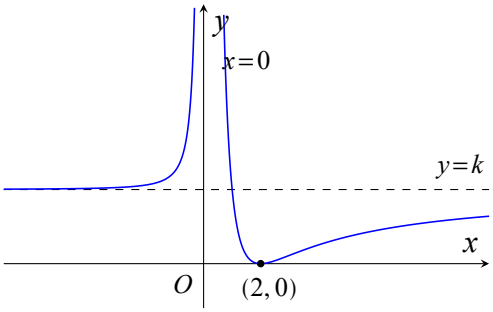
**C**: A reflection in the $y$-axis.

The resultant curve has equation $y=-\dfrac{x^2}{6x+45}$. Find $\mathrm{h}(x)$. [3]

**[NJC 2024 Promo Q4(b)]**

| Solution | | Comments |
|---|---|---|
| **[3]**<br><br>**(b)** | $y=-\dfrac{x^2}{6x+45}$<br>$\mathrm{C}^{-1}$: reflection in the $y$-axis $(x\to -x)$<br>$y=-\dfrac{(-x)^2}{6(-x)+45}=-\dfrac{x^2}{-6x+45}$<br>$\mathrm{B}^{-1}$: scaling parallel to the $x$-axis by factor $\frac{1}{3}$ $(x\to 3x)$<br>$y=-\dfrac{(3x)^2}{-6(3x)+45}=-\dfrac{9x^2}{-18x+45}=-\dfrac{x^2}{-2x+5}$<br>$\mathrm{A}^{-1}$: translation of 2 units in the negative $x$-direction $(x\to x+2)$<br>$y=-\dfrac{(x+2)^2}{-2(x+2)+5}=-\dfrac{(x+2)^2}{-2x+1}$<br>$\mathrm{h}(x)=\dfrac{(x+2)^2}{2x-1}$ | |

**4.2** The diagram below shows the graph of $y=\mathrm{f}(2x)$. The graph has a turning point at $(2,0)$, and asymptotes with equations $x=0$ and $y=k$.



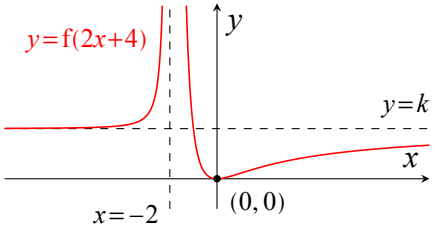
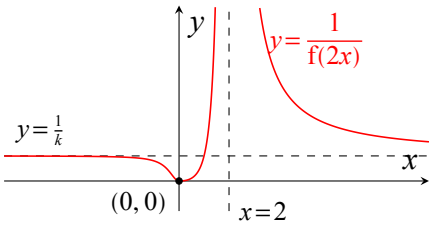
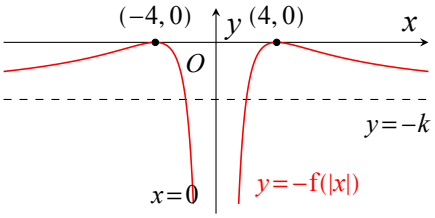

**(a)** State a single transformation that will transform the graph of $y=\mathrm{f}(2x)$ onto the graph of $y=\mathrm{f}(2x+4)$. Hence sketch the graph of $y=\mathrm{f}(2x+4)$. [3]

On separate clearly labelled diagrams, sketch the graphs of

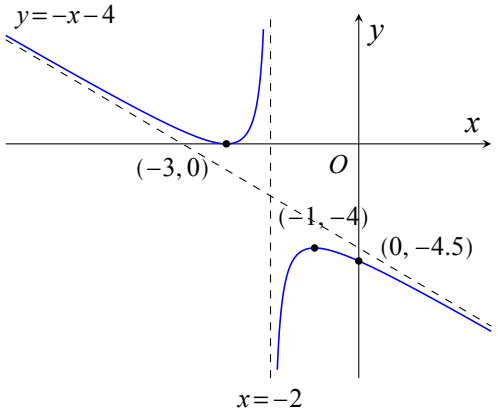
**(b)** $y=\dfrac{1}{\mathrm{f}(2x)}$, [3]

**(c)** $y=-\mathrm{f}(|x|)$. [3]

**[EJC 2025 Prelim P1 Q4]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[3]** |  | |
| **(b)**<br>**[3]** |  | |
| **(c)**<br>**[3]** |  | |

**4.3**




The diagram shows the curve $y = \mathrm{f}(x)$. The curve crosses the $x$-axis at its minimum point $(-3, 0)$. It has a maximum turning point $(-1, -4)$ and crosses the $y$-axis at the point $(0, -4.5)$. The asymptotes of the curve are $x = -2$ and $y = -x - 4$.

On separate diagrams, sketch the graphs of

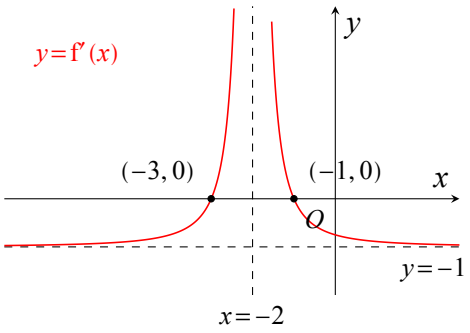
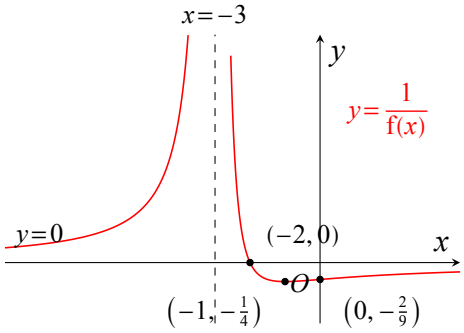
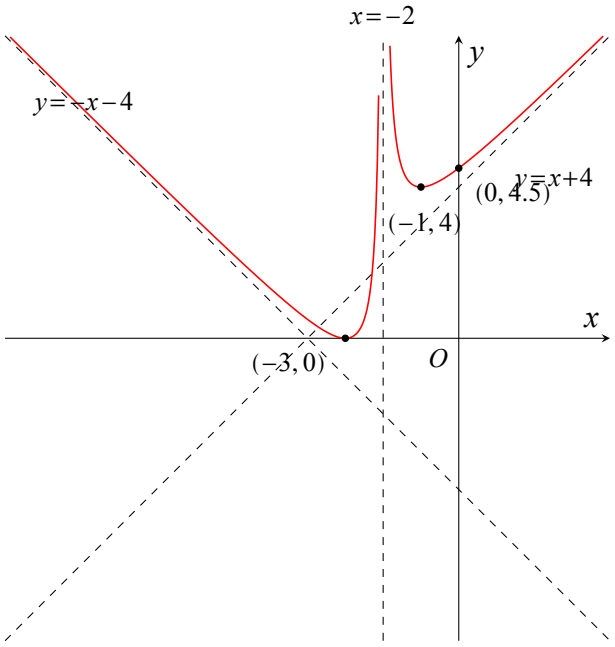
**(a)** $y = \mathrm{f}'(x)$, [3]

**(b)** $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

**(c)** $y = |\mathrm{f}(x)|$, [3]

indicating the equations of any asymptotes, turning points and intersections with the axes.

**[CJC 2025 Promo Q8]**

| Solution | Comments |
|---|---|
| **(a)** **[3]**   | |
| **(b)** **[3]**   | |
| **(c)** **[3]**   | |

**4.4** Express $\frac{9x+5}{3x+2}$ in the form $a-\frac{b}{3x+2}$ where $a$ and $b$ are positive constants.

Hence, describe a sequence of transformations which transform the graph of $y=\frac{9x+5}{3x+2}$ to the graph of $y=\frac{1}{x}$. [5]

**[RVHS 2026 Prelim P1 Q10(b)]**

| Solution | | Comments |
|---|---|---|
| **[5]**<br><br>**(b)** | $\frac{9x+5}{3x+2}=3-\frac{1}{3x+2} \quad (a=3,\ b=1)$<br>$y=3-\frac{1}{3x+2} \xrightarrow{y\to y+3} y=-\frac{1}{3x+2}$<br>$\xrightarrow{y\to -y} y=\frac{1}{3x+2}$<br>$\xrightarrow{x\to \frac{x}{3}} y=\frac{1}{x+2}$<br>$\xrightarrow{x\to x-2} y=\frac{1}{x}$<br><br>1. Translate 3 units in the negative $y$-direction<br>2. Reflect about the $x$-axis<br>3. Scale by factor 3 parallel to the $x$-axis<br>4. Translate 2 units in the positive $x$-direction | |

# 5 Functions

**5.1** The function f is defined by

$$\mathrm{f}: x \to \left|\frac{2x+4}{3-x}\right|, \qquad x \in \mathbb{R}, \qquad x \neq 3.$$

**(a)** Sketch the graph of $y = \mathrm{f}(x)$, giving the equations of any asymptotes and the coordinates of the points where the curve meets the axes. [3]

**(b)** Hence state the range of f. [1]

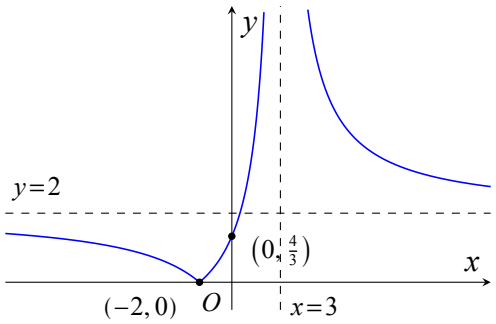
**(c)** Explain why the function $\mathrm{f}^2$ does not exist. [1]

The domain of f is further restricted to $x \leqslant a$, where $a$ is a constant.

**(d)** State the greatest value of $a$ such that the function $\mathrm{f}^{-1}$ exists. [1]

**(e)** Hence find $\mathrm{f}^{-1}(x)$ and state its domain. [4]

**[2023 A-Level P1 Q7]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $y = \left\|\frac{2x+4}{3-x}\right\| = \left\|-2+\frac{10}{3-x}\right\|$<br>Asymptotes: $y = \|-2\| = 2, \quad x = 3$<br>$x = 0: \quad y = \left\|\frac{4}{3}\right\| = \frac{4}{3}$<br>$y = 0: \quad x = -2$<br>  <br> | |
| **(b)** **[1]** | $\mathrm{R_f} = [0, \infty)$ | |

**(c)** **[1]**

$R_f = [0, \infty), \quad D_f = (-\infty, 3) \cup (3, \infty)$

$3 \in R_f,\ 3 \notin D_f \Rightarrow R_f \not\subseteq D_f$

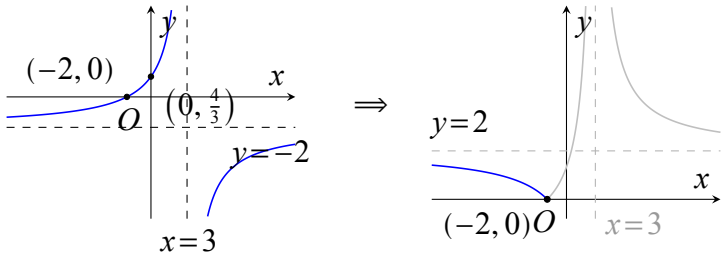
$\therefore f^2$ does not exist

**(d)** **[1]**

$a = -2$

**(e)** **[4]**

$x \leqslant -2: \quad f(x) = \left|\frac{2x+4}{3-x}\right| = -\frac{2x+4}{3-x}$




$y = f(x)$

$-3y + xy = 2x + 4$

$x(y-2) = 4 + 3y$

$x = \frac{4+3y}{y-2}$

$f^{-1}(x) = \frac{4+3x}{x-2}, \qquad D_{f^{-1}} = [0, 2)$

**5.2** The function f is defined by

$$\mathrm{f}: x \mapsto \frac{ax-3}{5x-4}, \text{ for } x \in \mathbb{R}, \ x \neq \frac{4}{5}, \text{ and } a \in \mathbb{R}.$$

**(a)** Find the range of f in terms of $a$. [1]

The function f is such that $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all $x$ in the domain of f.

**(b)** Find the value of $a$. [2]

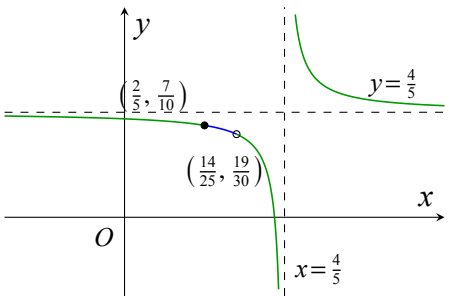
**(c)** Hence find $\mathrm{f}^{2025}(2)$. [2]

Another function g is defined by

$$\mathrm{g}: x \mapsto \left(x - \frac{1}{5}\right)^2 + \frac{2}{5}, \text{ for } x \in \mathbb{R}, \ 0 < x < \frac{3}{5}.$$

**(d)** Find the range of fg. [2]

**[HCI 2025 Prelim P1 Q8]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[1]** | Horizontal asymptote: $y=\frac{a}{5}$<br>$\mathrm{R_f}=\mathbb{R}\setminus\left\{\frac{a}{5}\right\}$ | |
| **(b)** **[2]** | $y=\frac{ax-3}{5x-4}$<br>$5xy-4y=ax-3$<br>$5xy-ax=4y-3$<br>$x=\frac{4y-3}{5y-a}$<br>$\mathrm{f}^{-1}(x)=\frac{4x-3}{5x-a},\quad x\in\mathbb{R},\ x\neq\frac{a}{5}$<br>$\mathrm{f}(x)=\mathrm{f}^{-1}(x)\ \Rightarrow\ a=4$ | |
| **(c)** **[2]** | $\mathrm{f}^{-1}(x)=\mathrm{f}(x)\ \Rightarrow\ \mathrm{f}^2(x)=\mathrm{ff}^{-1}(x)=x$<br>$\mathrm{f}^{2025}(2)=\mathrm{f}^2\left(\mathrm{f}^{2023}(2)\right)$<br>$=\mathrm{f}^2\left(\mathrm{f}^{2021}(2)\right)$<br>$=\cdots$<br>$=\mathrm{f}(2)$<br>$=\frac{4(2)-3}{5(2)-4}=\frac{5}{6}$ | |
| **(d)** **[2]** | $\mathrm{R_g}=\left[\frac{2}{5},\frac{14}{25}\right)$<br>$\mathrm{f}(x)=\frac{4x-3}{5x-4}$<br>$\left(0,\frac{3}{5}\right)\xrightarrow{\mathrm{g}}\left[\frac{2}{5},\frac{14}{25}\right)\xrightarrow{\mathrm{f}}\left(\frac{19}{30},\frac{7}{10}\right]$<br>$\mathrm{R_{fg}}=\left(\frac{19}{30},\frac{7}{10}\right]$<br>  <br> | |

**5.3** The function f is given by

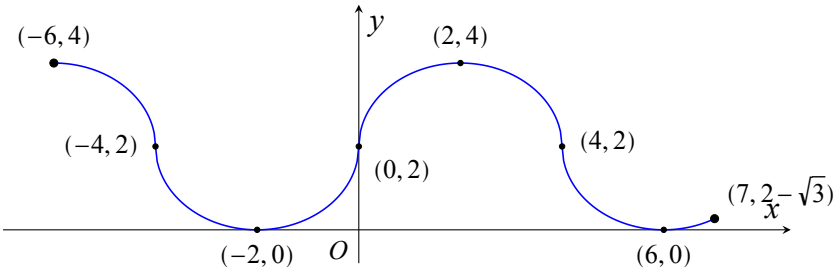
$$\mathrm{f}(x)=\begin{cases}2+\sqrt{2^2-(x-2)^2}, & 0<x\leqslant 4,\\ 2-\sqrt{2^2-(x-6)^2}, & 4<x\leqslant 8.\end{cases}$$

It is given that $\mathrm{f}(x)=\mathrm{f}(x+8)$ for all real values of $x$.

**(a)** On the diagram in the Printed Answer Book, sketch the graph of $y=\mathrm{f}(x)$ for $-6\leqslant x\leqslant 7$, indicating clearly the coordinates of the end points and the points where the graph cuts the axes. [3]

**(b)** Without integrating, write down the exact area of the region bounded by $y=\mathrm{f}(x)$, the line $x=4$, the $x$-axis and the $y$-axis. [1]

**[VJC 2025 Prelim P1 Q5(a), (b)]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $y=2+\sqrt{2^2-(x-2)^2}$<br>$y-2=\sqrt{2^2-(x-2)^2}$<br>$(x-2)^2+(y-2)^2=2^2$<br>Centre $(2,2)$, radius 2.<br>Since it is a positive square root, it is the top half of this circle.<br><br>$y=2-\sqrt{2^2-(x-6)^2}$<br>$y-2=-\sqrt{2^2-(x-6)^2}$<br>$(x-6)^2+(y-2)^2=2^2$<br>Centre $(6,2)$, radius 2.<br>Since it is a negative square root, it is the bottom half of this circle.<br><br>  <br> | |
| **(b)** **[1]** | $\text{Area}=2\times 4+\frac{1}{2}\pi(2^2)=8+2\pi$ | |

**5.4** The function f is defined by

$$\mathrm{f}(x) = 1 - ax, \ x \in \mathbb{R}, \text{ where } a \text{ is a real constant.}$$

The function g is defined by

$$\mathrm{g}(x) = \begin{cases} x^2 + 3 & \text{for} \quad 0 \leqslant x \leqslant 2, \\ 7 - x & \text{for} \quad 2 < x \leqslant 7. \end{cases}$$

**(i)** Find the set of possible values of $a$ such that $\mathrm{f}^{-1}$ exist. [1]

**(ii)** If $a = 2$, describe a sequence of transformations that transform the graph of $y = \mathrm{g}(x)$ onto the graph of $y = \mathrm{fg}(x)$. [3]

**(iii)** The function gg is defined by

$$\mathrm{gg}(x) = \begin{cases} \mathrm{h}_1(x) & \text{for} \quad 0 \leqslant x \leqslant 2, \\ \mathrm{h}_2(x) & \text{for} \quad 2 < x < 5, \\ \mathrm{h}_3(x) & \text{for} \quad 5 \leqslant x \leqslant 7. \end{cases}$$

Find $\mathrm{h}_1(x)$, $\mathrm{h}_2(x)$ and $\mathrm{h}_3(x)$. [3]

**[ACJC 2022 Prelim P2 Q1]**

| Solution | | Comments |
|---|---|---|
| **(i)**<br>**[1]** | $a \in \mathbb{R} \setminus \{0\}$ | |
| **(ii)**<br>**[3]** | $y = \mathrm{fg}(x) = 1 - 2\mathrm{g}(x)$<br>Reflection of the graph of $y = \mathrm{g}(x)$ in the $x$-axis, followed by scaling by scale factor 2 parallel to the $y$-axis, followed by translation 1 unit in the positive $y$-direction.<br>**OR**<br>Translation of the graph of $y = \mathrm{g}(x)$ by $\frac{1}{2}$ unit in the negative $y$-direction followed by reflection in the $x$-axis followed by scaling by scale factor 2 parallel to the $y$-axis.<br>**OR**<br>Translation of the graph of $y = \mathrm{g}(x)$ by $\frac{1}{2}$ unit in the negative $y$-direction followed by scaling by scale factor 2 parallel to the $y$-axis followed by reflection in the $x$-axis. | |
| **(iii)**<br>**[3]** | $\mathrm{gg}(x) = \begin{cases} 7-(x^2+3) & \text{for} \quad 0 \leqslant x \leqslant 2 \\ 7-(7-x) & \text{for} \quad 2 < x < 5 \\ (7-x)^2+3 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$<br>$\therefore \mathrm{gg}(x) = \begin{cases} 4-x^2 & \text{for} \quad 0 \leqslant x \leqslant 2 \\ x & \text{for} \quad 2 < x < 5 \\ x^2-14x+52 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$ | |

**5.5** Functions f and g are defined by

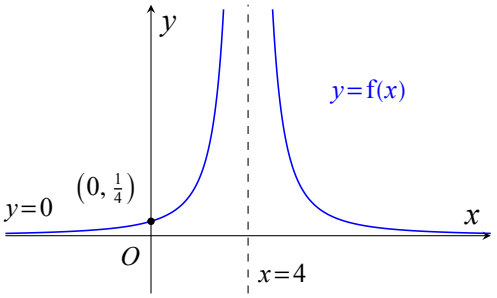
$$\mathrm{f}: x \to \frac{4}{(x-4)^2}, \quad x \in \mathbb{R},\ x \neq 4,$$

$$\mathrm{g}: x \to \ln\left(1+\frac{1}{x}\right), \quad x \in \mathbb{R},\ x > 0.$$

**(a)** Sketch the graph of $y = \mathrm{f}(x)$, stating the equations of any asymptotes, the coordinates of the points where it crosses the axes and the coordinates of the turning points, if any. [2]

**(b)** Show that gf exists. Hence find the rule, domain and range of gf. [4]

**[CJC 2025 Prelim P2 Q2(a), (b)]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[2]** |  | |

**(b)**
**[4]**

For gf to exist, $R_f \subseteq D_g$,

$R_f = (0, \infty)$

$D_g = (0, \infty)$

Since $R_f \subseteq D_g$, gf exists (shown)

$$\begin{aligned} \mathrm{gf}(x) &= \mathrm{g}\left[\frac{4}{(x-4)^2}\right] \\ &= \ln\left[1+\left(\frac{1}{\frac{4}{(x-4)^2}}\right)\right] \\ &= \ln\left[1+\frac{(x-4)^2}{4}\right] \end{aligned}$$

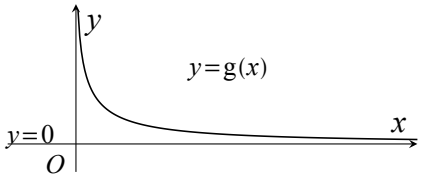
$D_{gf} = D_f = (-\infty, 4) \cup (4, \infty)$

OR

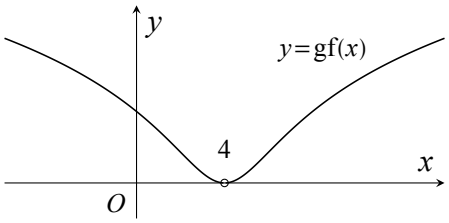
$D_{gf} = D_f = \mathbb{R} \setminus \{4\}$

**Method 1: Mapping**

$D_f = (-\infty, 4) \cup (4, \infty) \xrightarrow{f} R_f = (0, \infty) \xrightarrow{g} R_{gf} = (0, \infty)$




**Method 2: Sketch $y = \mathrm{gf}(x)$ for given domain**




$R_{gf} = (0, \infty)$

**5.6** The function g is defined by $g(0)=g(1)=1$, $g(2n)=2+g(n)$ and $g(2n+1)=1+g(2n)$, for integer $n$.

**(i)** Find $g(4)$, $g(7)$ and $g(12)$.

**(ii)** Does g have an inverse? Justify your answer. [3]

**[2016 A-Level P1 Q10(b)]**

| **Solution** | | **Comments** |
|---|---|---|
| **(i)** | $g(4)=2+g(2)$<br>$=2+2+g(1)$<br>$=2+2+1+g(0)$<br>$=2+2+1+1$<br>$=6$<br><br>$g(7)=1+g(6)$<br>$=1+2+g(3)$<br>$=1+2+1+g(2)$<br>$=1+2+1+2+1+1$<br>$=8$<br><br>$g(12)=2+g(6)$<br>$=2+2+1+2+1+1$<br>$=9$ | |
| **(ii)**<br>**[3]** | $g(8)=2+g(4)$<br>$=2+6$<br>$=8$<br>$=g(7)$<br>Since $g(8)=g(7)=8$, g is not one-one, so $g^{-1}$ does not exist. | |

# 6 Sequences and Series

**6.1** It is given $\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$ and $u_r = 2r^3 + 5$.

**(i)** Find $\sum_{r=1}^{n} u_r$. [2]

**(ii)** Hence, or otherwise, find $\sum_{r=2}^{n} (2(r+2)^3 + 5)$. [3]

**(iii)** Explain why the series $\sum_{r=1}^{\infty} u_r$ does not converge. [1]

**[JPJC 2025 Prelim P1 Q8(b)]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[2]** | $\sum_{r=1}^{n} (2r^3+5) = 2\sum_{r=1}^{n} r^3 + \sum_{r=1}^{n} 5$<br>$= 2\left(\frac{n^2(n+1)^2}{4}\right) + 5n$<br>$= \frac{n^2(n+1)^2}{2} + 5n$ | |
| **(ii)** **[3]** | $\sum_{r=2}^{n} (2(r+2)^3+5) = (2(2+2)^3+5) + \cdots + (2(n+2)^3+5)$<br>$= \sum_{r=4}^{n+2} (2r^3+5)$<br>$= \sum_{r=1}^{n+2} (2r^3+5) - \sum_{r=1}^{3} (2r^3+5)$<br>$= \frac{(n+2)^2(n+3)^2}{2} + 5(n+2) - \left[\left(\frac{3^2(4)^2}{2}\right) + 5(3)\right]$<br>$= \frac{(n+2)^2(n+3)^2}{2} + 5n - 77$ | |
| **(iii)** **[1]** | From (i) $\sum_{r=1}^{n} u_r = \frac{n^2(n+1)^2}{2} + 5n$<br>As $n \to \infty$, $\frac{n^2(n+1)^2}{2} \to \infty$, $5n \to \infty$<br>Hence, $\sum_{r=1}^{n} u_r = \frac{n^2(n+1)^2}{2} + 5n \to \infty$, series $\sum_{r=1}^{\infty} u_r$ does not converge. | |

**6.2** It is given that $\sum_{r=2}^{n} \frac{1}{r(r^2-1)} = \frac{1}{4} + \frac{1}{2(n+1)} - \frac{1}{2n}$.

**(a)** Show that $\sum_{r=2}^{n} \frac{1}{r(r^2-1)}$ is less than $\frac{1}{4}$. [2]

**(b)** Give a reason why the series $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)}$ converges, and write down its value. [2]

**(d)** Find $\sum_{r=m+1}^{N} \frac{1}{(r-m)(r-m+1)(r-m+2)}$, where $m$ and $N$ are integers with $N > m > 0$.
(There is no need to express your answer as a single algebraic fraction.) [2]

**[VJC 2025 Prelim P1 Q6(a), (b), (d)]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | $\frac{1}{2(n+1)} - \frac{1}{2n} = \frac{n-(n+1)}{2n(n+1)} = \frac{-1}{2n(n+1)} < 0$ for all $n \geqslant 2$<br>$\sum_{r=2}^{n} \frac{1}{r(r^2-1)} = \frac{1}{4} + \frac{1}{2(n+1)} - \frac{1}{2n} < \frac{1}{4}$ | |
| **(b)** **[2]** | As $n \to \infty$, $\frac{1}{2(n+1)} \to 0$ and $\frac{1}{2n} \to 0$,<br>$\therefore \sum_{r=2}^{n} \frac{1}{r(r^2-1)} \to \frac{1}{4}$, a constant.<br>Hence $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)}$ converges and $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)} = \frac{1}{4}$. | |
| **(d)** **[2]** | $\sum_{r=m+1}^{N} \frac{1}{(r-m)(r-m+1)(r-m+2)}$<br>$= \frac{1}{1(2)(3)} + \frac{1}{2(3)(4)} + \cdots + \frac{1}{(N-m)(N-m+1)(N-m+2)}$<br>$= \sum_{r=2}^{N-m+1} \frac{1}{(r-1)(r)(r+1)}$<br>$= \sum_{r=2}^{N-m+1} \frac{1}{r(r^2-1)}$<br>$= \frac{1}{4} + \frac{1}{2(N-m+2)} - \frac{1}{2(N-m+1)}$ | |

**6.3** A sequence is such that $u_{n+1} = u_n + 1.5$, for $n = 1, 2, 3, \ldots$.

**(a)** Given that $u_{50} = 99u_1$, show that $u_1 = 0.75$. [2]

**(b)** Find the least value of $n$ such that $u_2 + u_4 + u_6 + \cdots + u_{2n}$ is more than 2025. [2]

**(c)** It is given that $u_{25}$, $u_k$ and $u_5$ are the first three terms of a geometric progression.

**(i)** Find $k$. [2]

**(ii)** Find the sum of the first 13 terms of the geometric series, giving your answer to 3 decimal places. [2]

**[NYJC 2025 Prelim P1 Q8]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | Note that the sequence is an arithmetic progression with common difference 1.5<br>$u_{50} = u_1 + 49(1.5)$<br>$99u_1 = u_1 + 73.5$<br>$u_1 = 0.75$ | |
| **(b)** **[2]** | **Method 1**<br>$\sum_{k=1}^{n} u_{2k} = u_2 + u_4 + \cdots + u_{2n}$ is a sum of $n$ terms of an arithmetic sequence with first term $u_2 = 0.75 + 1.5 = 2.25$ and common difference $2(1.5) = 3$<br>$u_2 + u_4 + \cdots + u_{2n} = \frac{n}{2}(2(2.25) + (n-1)(3))$<br>$= \frac{n}{2}(1.5 + 3n)$<br>From GC, for $\frac{n}{2}(1.5+3n) > 2025$, least $n = 37$<br>**Method 2**<br>$\sum_{k=1}^{n} u_{2k} = \sum_{k=1}^{n} (2.25 + (2k-1)3) > 2025$<br>Using GC, least $n = 37$ | |
| **(c)(i)** **[2]** | $\frac{u_k}{u_{25}} = \frac{u_5}{u_k}$<br>$u_k^2 = u_5 u_{25}$<br>$[0.75 + (k-1)(1.5)]^2 = [0.75 + (5-1)(1.5)]\,[0.75 + (25-1)(1.5)]$<br>$0.75 + (k-1)(1.5) = \sqrt{6.75 \times 36.75} = 15.75$<br>$k = 11$ | |
| **(c)(ii)** **[2]** | **Method 1**<br>Common ratio $= \frac{15.75}{36.75} = \frac{3}{7}$<br>$S_{13} = \frac{36.75\left(1 - \left(\frac{3}{7}\right)^{13}\right)}{1 - \frac{3}{7}}$<br>$= 64.31144$<br>$= 64.311$<br>**Method 2**<br>Using GC, $\sum_{r=1}^{13} \left(36.75\left(\frac{3}{7}\right)^{r-1}\right) = 64.311$ | |

**6.4** A sequence $u_1, u_2, u_3, \ldots$ is defined by

$$u_{n+1} = 2u_n\,(1-u_n)$$

for all positive integers $n$.

**(a)** Find the possible limits of the sequence, if the sequence is convergent. [2]

**(b)** For each of the following, evaluate $u_2$ and $u_3$, and state whether the sequence is convergent:

- $u_1 = 0$,
- $u_1 = 0.01$,
- $u_1 = -0.01$. [6]

**[EJC 2025 Prelim P2 Q2]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | Let $L$ be the limit of the sequence $u_{n+1} = 2u_n\,(1-u_n)$.<br>$L = 2L(1-L)$<br>$2L^2 - L = 0$<br>$L(2L-1) = 0$<br>$L = 0$ or $L = \frac{1}{2}$ | |
| **(b)** **[6]** | When $u_1 = 0$, $u_2 = 0$, $u_3 = 0$, sequence is convergent.<br>When $u_1 = 0.01$, $u_2 = 0.0198$ (exact), $u_3 = 0.0388$ (3 s.f.), sequence is convergent.<br>When $u_1 = -0.01$, $u_2 = -0.0202$ (exact), $u_3 = -0.0412$ (3 s.f.), sequence is not convergent. | |

# 7 Arithmetic and Geometric Progressions

**7.1** **(a)** **(i)** The ninth, fifth and second terms of an arithmetic progression are successive terms of a geometric progression with first term $a$ and common ratio $r$, where $r \neq 1$ and $a > 0$. Find the value of $r$ and deduce that the geometric series is convergent. [3]

**(ii)** Using the value of $r$ found in (i), find the least value of $n$ such that the sum of all the terms after the $n$th term of the geometric progression is less than 1% of its sum of the first $n$ terms. [3]

**(b)** The sum, $S_n$, of the first $n$ terms of another sequence is given by $S_n = n \ln 2^{q(n+1)}$, where $q$ is a constant. Prove that the sequence follows an arithmetic progression. [4]

**[ASRJC 2025 Prelim P1 Q8]**

| Solution | | Comments |
|---|---|---|
| **(a)(i)** **[3]** | Let the first term and common difference of the arithmetic progression be $b$ and $d$ respectively.<br>Since $b+8d$, $b+4d$ and $b+d$ are successive terms of a GP, then $\frac{b+4d}{b+8d}=\frac{b+d}{b+4d}$ (common ratio).<br>$(b+4d)^2=(b+8d)(b+d)$<br>$8bd+16d^2=9bd+8d^2$<br>$d(8d-b)=0$<br>$b=8d \quad (d\neq 0)$<br>$\Rightarrow r=\frac{8d+4d}{8d+8d}=\frac{3}{4}$<br>Since $\lvert r\rvert=\left\lvert\frac{3}{4}\right\rvert<1$, the series is convergent. | |
| **(a)(ii)** **[3]** | $S_\infty - S_n < 0.01S_n \Rightarrow S_\infty < 1.01S_n$<br>$\frac{a}{1-\frac{3}{4}} < 1.01\left[\frac{a\left(1-\left(\frac{3}{4}\right)^n\right)}{1-\frac{3}{4}}\right]$<br>$1<1.01\left(1-\left(\frac{3}{4}\right)^n\right)$<br>$\left(\frac{3}{4}\right)^n<0.009901$<br>$n\ln\left(\frac{3}{4}\right)<\ln(0.009901)$<br>$n>\frac{\ln(0.009901)}{\ln\left(\frac{3}{4}\right)}=16.042$<br>Hence least $n$ is 17. | |
| **(b)** **[4]** | $T_n=S_n-S_{n-1}=n\ln 2^{q(n+1)}-(n-1)\ln 2^{q(n)}$<br>$=n(n+1)q\ln 2-(n-1)(nq)\ln 2$<br>$=nq\ln 2+nq\ln 2$<br>$=2nq\ln 2$<br>$T_n-T_{n-1}=2nq\ln 2-2(n-1)q\ln 2$<br>$=2q\ln 2 \quad$ (constant $\because q$ is a constant)<br>$\therefore$ the sequence is an arithmetic progression (Proved). | |

**7.2** An arithmetic series has first term $a$ and common difference $d$, where $a$ and $d$ are non-zero real constants. A convergent geometric series has first term $b$ and common ratio $r$, where $b>0$ and $r$ is a non-zero real constant. The sum of the first $n$ terms of the geometric series is denoted by $S$.

It is given that the 7th, 10th and 11th terms of the arithmetic series are equal to the 3rd, 6th and 10th terms of the geometric series respectively.

**(a)** Show that $r$ satisfies the equation $3r^7 - 4r^3 + 1 = 0$. Solve this equation, giving your answer correct to 4 decimal places. [4]

**(b)** A new series is formed by taking the even numbered terms of the geometric series. Find the smallest value of $n$ for which $S$ differs from the sum to infinity of the new series by at least $1.78b$. [4]

**[TMJC 2025 Prelim P1 Q4]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[4]** | $a+6d=br^2 \quad --(1)$<br>$a+9d=br^5 \quad --(2)$<br>$a+10d=br^9 \quad --(3)$<br>$(2)-(1): 3d=br^2(r^3-1) \quad --(4)$<br>$(3)-(2): d=br^5(r^4-1) \quad --(5)$<br>Substitute (5) into (4),<br>$3br^5(r^4-1)=br^2(r^3-1)$<br>$3r^3(r^4-1)=(r^3-1) \quad$ (since $b, r\neq 0$)<br>$3r^7-4r^3+1=0 \quad$ (shown)<br>By GC, since the series is convergent, $\lvert r\rvert<1$,<br>$r=0.6639$ (4 d.p.)<br>[GC SCREEN PLACEHOLDER] GC 1: [APPS] PlySmlt2, POLY ROOT FINDER, ORDER 7 [NEXT]<br>[GC SCREEN PLACEHOLDER] GC 2: coefficients $a_7=3,\ a_6=0,\ a_5=0,\ a_4=0,\ a_3=-4,\ a_2=0,\ a_1=0,\ a_0=1$ [SOLVE] | |
| **(b)** **[4]** | New Series: $br, br^3, br^5, \ldots$<br>$\left\lvert \frac{br}{1-r^2}-S\right\rvert \geqslant 1.78b$<br>$\left\lvert \frac{br}{1-r^2}-\frac{b}{1-r}(1-r^n)\right\rvert \geqslant 1.78b$<br>$\left\lvert \frac{0.6639b}{1-0.6639^2}-\frac{b}{1-0.6639}(1-0.6639^n)\right\rvert \geqslant 1.78b$<br>$\left\lvert \frac{0.6639}{1-0.6639^2}-\frac{1}{1-0.6639}(1-0.6639^n)\right\rvert \geqslant 1.78 \quad (\because b>0)$<br>Using GC,<br>When $n=14$, $\left\lvert \frac{0.6639}{1-0.6639^2}-\frac{1}{1-0.6639}(1-0.6639^n)\right\rvert = 1.7785 < 1.78$<br>When $n=15$, $\left\lvert \frac{0.6639}{1-0.6639^2}-\frac{1}{1-0.6639}(1-0.6639^n)\right\rvert = 1.7818 \geqslant 1.78$<br>Therefore, least $n=15$. | |

**7.3** An arithmetic progression has first term $a$ and common difference $d$, where $a$ and $d$ are non-zero. The 3rd, 9th and 11th terms of the arithmetic progression are three distinct consecutive terms of a geometric progression.

**(a)** Find $d$ in terms of $a$. [3]

**(b)** Let the sum of the first $n$ terms of the arithmetic progression be $S_n$ where $S_1 = 143$. Find the largest possible value of $S_n$. [3]

The first term of the geometric progression is $b$, where $b$ is positive.

**(c)** Determine whether the geometric series is convergent. [2]

**(d)** Show that the sum of all terms after the $n$th term of the geometric progression is at most $\frac{1}{2}b$, where $n$ is a positive integer. [2]

**[SAJC 2025 Prelim P1 Q5]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)**<br>**[3]** | $\frac{a+8d}{a+2d} = \frac{a+10d}{a+8d}$<br>$(a+8d)^2 = (a+2d)(a+10d)$<br>$a^2+16ad+64d^2 = a^2+12ad+20d^2$<br>$4ad+44d^2 = 0$<br>$4d(a+11d) = 0$<br>$d=0$ (reject since terms of AP, GP are distinct) or $a+11d=0$<br>$d = -\frac{1}{11}a$ | |

**(b)** **[3]**

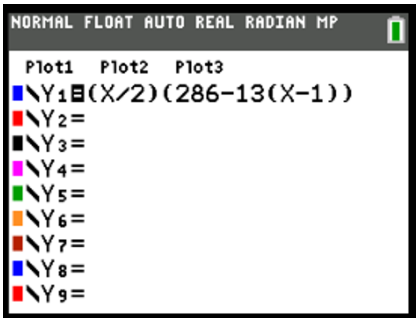
$S_1 = a = 143$

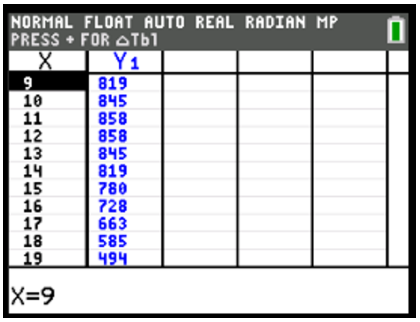
$$d = -\frac{1}{11}(143) = -13$$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$= \frac{n}{2}[286 - 13(n-1)]$$

From GC, largest $S_n = 858$




**(c)** **[2]**

$$r = \frac{a+8d}{a+2d} = \frac{-11d+8d}{-11d+2d} = \frac{-3d}{-9d} = \frac{1}{3}$$

Since $|r| = \frac{1}{3} < 1$, the geometric series is convergent.

**(d)** **[2]**

Let $u_n$ be the $n$th term of the geometric progression.

$u_{n+1} + u_{n+2} + \ldots$

$= S_\infty - S_n$

$$= \frac{b}{1-r} - \frac{b(1-r^n)}{1-r}$$

$$= \frac{br^n}{1-r}$$

$$= \frac{b\left(\frac{1}{3}\right)^n}{1-\frac{1}{3}}$$

$$= \frac{3}{2}\left(\frac{1}{3}\right)^n b$$

$$= \frac{b}{2}\left(\frac{1}{3}\right)^{n-1} \leqslant \frac{1}{2}b \quad \text{since } \left(\frac{1}{3}\right)^{n-1} \leqslant 1$$

Alternative:

$u_{n+1} + u_{n+2} + u_{n+3} + \ldots$

$$= b\left(\frac{1}{3}\right)^n + b\left(\frac{1}{3}\right)^{n+1} + b\left(\frac{1}{3}\right)^{n+2} + \ldots$$

$$= \frac{b\left(\frac{1}{3}\right)^n}{1-\frac{1}{3}}$$

$$= \frac{3}{2}b\left(\frac{1}{3}\right)^n$$

$$= \frac{b}{2}\left(\frac{1}{3}\right)^{n-1} \leqslant \frac{b}{2} \quad \text{since } \left(\frac{1}{3}\right)^{n-1} \leqslant 1$$

**7.4** Mabel and Janice decided to start a 5-year savings plan beginning in January 2026.

Mabel saves using a piggy bank. At the start of January 2026, she deposits \$101. Each subsequent month, she increases her deposit by \$1—so she deposits \$101 in January, \$102 in February, \$103 in March, and so on, until \$112 in December. At the start of each new year, she resets her monthly deposit to \$101 in January and repeats the same pattern through December. She continues this routine from 2026 to 2030, inclusive.

Janice, on the other hand, deposits \$100 at the start of every month into a bank account that earns 0.3% interest per month, with interest calculated and added into the account at the end of each month.

**(a)** Show that Janice will have more money in her savings account than Mabel has in her piggy bank at the end of December 2030. [5]

**(b)** Find the month and year when Janice's savings first exceed Mabel's savings. [4]

**[TJC 2025 Prelim P1 Q7]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[5]** | For Mabel:<br>Amount of money Mabel saves in a year<br>$=\$(101+102+\cdots+112)$<br>$=\$\frac{12}{2}(101+112)=\$1278$<br>Amount of money Mabel saves in 5 year $=\$1278\times 5=\$6390$<br><br>For Janice: | |

| $n$th mth | Amount at start of month ($) | Amount at end of month ($) |
|---|---|---|
| 1 Jan26 | 100 | $100(1.003)$ |
| 2 | $100+100(1.003)$<br>$=100(1+1.003)$ | $100(1+1.003)(1.003)$<br>$=100\,(1.003+1.003^2)$ |
| ⋮ | | ⋮ |
| $n$ | | $100\,(1.003+1.003^2+\cdots+1.003^n)$<br>$=\frac{100(1.003)\,(1.003^n-1)}{1.003-1}$ |

At Dec 2030, $n=60$.

Therefore, Amount Janice saves in 5 years

$=\frac{100(1.003)\,(1.003^{60}-1)}{1.003-1}=\$6582.85$ (2 d.p.) [A1]

$\therefore$ Janice will have more money in her savings account than Mabel has in her piggy bank.

**(b)**
**[4]**

At end of 4th year,

Mabel will have \$1278 x 4 $=\$5112$

Janice will have $\frac{100(1.003)\,(1.003^{48}-1)}{1.003-1}=\$5169.97>\$5112$

At end of 3rd year,

Mabel will have \$1278 x 3 $=\$3834$

Janice will have $\frac{100(1.003)\,(1.003^{36}-1)}{1.003-1}=\$3806.97<\$3834$

Therefore, the year when Janice's saving is more than Mabel's saving happen in 4th year (i.e. 2029). Let it be the $k^{\text{th}}$ month.

Then set Janice's saving > Mabel's saving

$\Rightarrow \frac{100(1.003)\,(1.003^{36+k}-1)}{1.003-1}>3834+\frac{k}{2}\,[2(100)+(k-1)(1)]$

$\Rightarrow \left(\frac{100003}{3}\right)(1.003^{36+k}-1)-3834-\frac{k}{2}\,[199+k]>0$

Using GC, we have $k\geqslant 4$. Therefore, the month and year is Apr 2029.

**7.5** Sarah, a Singaporean Citizen, started a 4-year undergraduate Computing course at the National University of Singapore (NUS) in August 2021. The annual tuition fee, after the MOE Tuition Grant, is \$8,250. Sarah took a Tuition Fee Loan (TFL) covering 90% of her tuition fees. This loan is interest-free during her course of study.

Sarah decided to set aside \$$x$ per month from August 2021. This amount is increased by \$2 each subsequent month. (Sarah set aside \$$x$ in August 2021, \$$(x+2)$ in September 2021, ….) With the amount set aside, she repaid one-fifth of her total TFL at the end of July 2025.

**(a)** Find the value of $x$. [3]

Upon Sarah's graduation, the TFL's interest rate becomes effective at a fixed rate of 0.4% per month, calculated on the outstanding balance at the end of the month. The interest is first charged on 31st August 2025.

Sarah plans to make monthly repayments of \$$y$ to the bank on the 1st of every month, starting from 1st September 2025.

**(b)** Find the amount of outstanding loan after Sarah's first monthly repayment. [1]

**(c)** Show that the amount of outstanding loan after Sarah's $n$th monthly repayment is

$$23760(1.004)^n - 250y\,(1.004^n - 1)\,.$$

[3]

**(d)** Sarah wants to pay off her remaining loan by the end of 3 years. Find the value of $y$. [2]

Sarah decided on a different repayment strategy. For the first 24 months, she repays \$500 per month. After these 24 months, she doubles her monthly repayment to \$1000.

**(e)** Determine the month and year Sarah will clear her loan and calculate the amount of her final repayment. [4]

**[ACJC 2025 Prelim P1 Q12]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[3]** | Amount Sarah plans to repay (before graduation):<br>$\frac{1}{5}\times\frac{90}{100}\times 8250\times 4=\$5940$<br>Number of repayments made from August 2021 to July 2025 is 48 (i.e. $n=48$).<br>The repayments follow an arithmetic progression (AP) with first term $a=x$ and common difference $d=2$.<br>The sum of an arithmetic progression is<br>$S_n=\frac{n}{2}[2a+(n-1)d]$<br>$5940=\frac{48}{2}[2x+(48-1)2]$<br>$x=76.75$ | |
| **(b)**<br>**[1]** | Amount of outstanding loan after Sarah's graduation:<br>$\frac{4}{5}\times\frac{90}{100}\times 8250\times 4=\$23760$<br>On 31st August 2025, apply interest of 0.4% to outstanding loan to get $\$(23760(1.004))$.<br>On 1st September 2025, make 1st repayment of $\$y$ to the bank.<br>$\therefore$ Outstanding loan after 1st repayment<br>$=\$(23760(1.004)-y)$ or $\$(23855.04-y)$ | |
| **(c)**<br>**[3]** | On 30th September 2025, apply interest of 0.4%.<br>On 1st October 2025, make 2nd repayment of $\$y$.<br>Outstanding loan after 2nd repayment<br>$=(23760(1.004)-y)\ 1.004-y$<br>$=23760(1.004)^2-1.004y-y$<br>Outstanding loan after 3rd repayment (on 1st Nov 2025)<br>$=(23760(1.004)^2-1.004y-y)\ 1.004-y$<br>$=23760(1.004)^3-1.004^2y-1.004y-y$<br>Outstanding loan after $n^{th}$ repayment<br>$=23760(1.004)^n-1.004^{n-1}y-\cdots-1.004^2y-1.004y-y$<br>$=23760(1.004)^n-y\,(1+1.004+1.004^2+\cdots+1.004^{n-1})$<br>$=23760(1.004)^n-y\left(\frac{1.004^n-1}{1.004-1}\right)$<br>$=23760(1.004)^n-250y\,(1.004^n-1)$ (shown) | |

**(d)** **[2]**

To repay remaining loan by the end of 3 years (i.e. 36 monthly repayments of $\$y$), let $n=36$.

To fully repay the loan, we want outstanding loan amount after 36 repayments to be zero (or less).

$$23760(1.004)^{36}-250y\,(1.004^{36}-1)\leqslant 0$$

$$y\geqslant 709.9769$$

$$\therefore y=709.98 \text{ (nearest cent)}$$

**(e)** **[4]**

After 24 monthly repayments of \$500 (i.e. $n=24$, $y=500$), outstanding loan amount

$$=23760(1.004)^{24}-250(500)\,((1.004)^{24}-1)$$

$$=13580.49$$

Outstanding loan amount as of 1st August 2027

$$=13580.49$$

In a similar method to **(c)**, the outstanding loan amount after $m$ monthly repayments of \$1000 (i.e. $y=1000$) is

$$13580.49(1.004)^{m}-250(1000)\,((1.004)^{m}-1)$$

To fully repay the loan, we need

$$13580.49(1.004)^{m}-250(1000)\,((1.004)^{m}-1)\leqslant 0$$

$$m\geqslant 13.991185$$

$$\text{Least } m=14$$

Sarah would require another 14 monthly repayments to fully repay her loan, with her first repayment of \$1000 on 1st September 2027.

Hence, she would clear her loan on (1st) October 2028.

The outstanding loan amount after her 13th monthly repayment of \$1000 (on 1st Sep 2028) is

$$13580.49(1.004)^{13}-250(1000)\,((1.004)^{13}-1)$$

$$=987.2529$$

Interest is applied on 30th Sep 2028. Hence her final repayment amount is

$$987.2529\times 1.004=\$991.20 \text{ (nearest cent)}$$

(Alternatively, outstanding loan amount after 14th monthly repayment is

$$13580.49(1.004)^{14}-250(1000)\,((1.004)^{14}-1)$$

$$=-8.7981$$

Hence the final repayment is $1000-8.7981=\$991.20$ (nearest cent).

**7.6** The dosage of medicine given to a patient needs to be carefully managed in order to achieve the desired result.

**(a)** To give the patient time to adapt to a particular medicine, the dosage is slowly increased over time. The patient is initially given a dosage of 0.75 units. The dosage is increased by 0.5 units each day until a total of at least 10 units of medicine has been taken. Find the least number of days it takes to achieve this, and the dosage on that day. [4]

**(b)** The doctor prescribed a new medication to another patient. After $n$ days, the total dosage of the new medicine administered to the patient is given by

$$\frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right) \text{ units.}$$

**(i)** Show that the increase in dosage of the new medicine administered per day follows a geometric progression. [3]

**(ii)** Find the total dosage of the new medicine administered to the patient after a long period of treatment. [2]

**(c)** Doctors can manage the medication so that the daily dosage of medicine increases as an arithmetic progression. Explain in context why the arithmetic progression is not a preferred model for the dosage of medication. [1]

**[RVHS 2025 Prelim P2 Q4]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[4]** | Let $u_n$ be the dose on the $n^{th}$ day.<br>$u_n = 0.75 + 0.5(n-1)$<br>$S_n = \frac{n}{2}(2(0.75) + (n-1)(0.5))$<br>$10 \leqslant \frac{n}{2}\left(1 + \frac{n}{2}\right)$ ——$(*)$<br>$\frac{1}{4}n^2 + \frac{1}{2}n \geqslant 10$<br>$n^2 + 2n \geqslant 40$<br>$(n+1)^2 \geqslant 41$<br>$n \geqslant -1 + \sqrt{41}$ or $n \leqslant -1 - \sqrt{41}$<br>(rej as no. of days cannot be -ve)<br>$n \geqslant 5.40312$<br>Therefore, it takes 6 days.<br>Alternatively,<br>When $n = 5$, $\frac{1}{4}n^2 + \frac{1}{2}n = 8.75 < 10$<br>When $n = 6$, $\frac{1}{4}n^2 + \frac{1}{2}n = 12 \geqslant 10$<br>Therefore, it takes 6 days.<br>$u_6 = 0.75 + 0.5(5) = 3.25$<br>The dose on the 6th day is 3.25 units. | |

| | | |
|---|---|---|
| **(b)(i)**<br>**[3]** | Increase on $n^{th}$ day<br>$= S_n - S_{n-1}$<br>$= \frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right) - \frac{3}{20}\left(\frac{5^{n-1} - 4^{n-1}}{5^{n-2}}\right)$<br>$= \frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}} - \frac{5^{n-1} - 4^{n-1}}{5^{n-2}}\right)$<br>$= \frac{3}{20}\left(\frac{1}{5^{n-2}}\right)\left(5^{n-1} - \frac{4}{5}(4^{n-1}) - 5^{n-1} + 4^{n-1}\right)$<br>$= \frac{3}{20}\left(\frac{1}{5^{n-2}}\right)\left(\frac{1}{5}\right)(4^{n-1})$<br>$= \frac{3}{20}\left(\frac{4}{5}\right)^{n-1}$<br>which is of the form $ar^{n-1}$, where $a = \frac{3}{20}$ and $r = \frac{4}{5}$ which is a GP.<br>**Alternative:**<br>Consider $\frac{u_{n+1}}{u_n} = \frac{\frac{3}{20}\left(\frac{4}{5}\right)^{n+1-1}}{\frac{3}{20}\left(\frac{4}{5}\right)^{n-1}} = \frac{4}{5}$ which is a constant<br>Since consecutive terms have a common ratio $r = \frac{4}{5}$, the increase in dosage follows a GP. | |

| | | |
|---|---|---|
| **(b)(ii)**<br>**[2]** | **Method 1**<br>Long term dosage<br>$=\lim_{n\to\infty}\frac{3}{20}\left(\frac{5^n-4^n}{5^{n-1}}\right)$<br>$=\frac{3/20}{1-4/5}$<br>$=\frac{3}{4}$<br>In the long term, the total dosage approaches 0.75 units.<br>**Note:**<br>After $n$ days, the total dosage of the medicine is given by $\frac{3}{20}\left(\frac{5^n-4^n}{5^{n-1}}\right)$.<br>Let $S_n$ denote the total dosage of the new medicine after $n$ days.<br>Then $S_n=\frac{3}{20}\left(\frac{5^n-4^n}{5^{n-1}}\right)$.<br>Let $v_n$ the change in dosage on $n$th day found in part (b)(i) where $v_n=S_n-S_{n-1}$<br>Since $v_n$ is a geometric progression, we can use $S_\infty=\frac{a}{1-r}$, the sum to infinity formula to find the long term total dosage.<br>**Method 2**<br>$\frac{3}{20}\left(\frac{5^n-4^n}{5^{n-1}}\right)$<br>$=\frac{3}{20}(5)\left(\frac{5^n-4^n}{5^n}\right)$<br>$=\frac{3}{20}(5)\left(1-\left(\frac{4}{5}\right)^n\right)$<br>As $n\to\infty$, $\left(\frac{4}{5}\right)^n\to 0$<br>Total dosage in the long term<br>$=\frac{3}{20}(5)(1-0)$<br>$=\frac{3}{4}$ | |
| **(c)**<br>**[1]** | If the dosage increases as an AP, then it will continue to **increase to** (positive) **infinity**. | |

# 8 Differentiation

**8.1** A curve $C$ has equation $x^2+y^2-4y=xy$, where $x \geqslant 0$.

**(a)** Show that $(2y-x-4)\dfrac{\mathrm{d}y}{\mathrm{d}x}=y-2x$. [2]

**[RVHS 2025 Prelim P1 Q7(a)]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[2]** | $x^2+y^2-4y=xy$<br>$2x+2y\dfrac{\mathrm{d}y}{\mathrm{d}x}-4\dfrac{\mathrm{d}y}{\mathrm{d}x}=y+x\dfrac{\mathrm{d}y}{\mathrm{d}x}$<br>$(2y-x-4)\dfrac{\mathrm{d}y}{\mathrm{d}x}+2x-y=0$<br>$(2y-x-4)\dfrac{\mathrm{d}y}{\mathrm{d}x}=y-2x$ | |

**8.2** A curve has parametric equations $x=t+\sin t$, $y=1-\cos t$.

Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\tan\dfrac{t}{2}$. [3]

| Solution | | Comments |
|---|---|---|
| **[3]** | $x=t+\sin t \Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}t}=1+\cos t$<br>$y=1-\cos t \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}t}=\sin t$<br>$\dfrac{\mathrm{d}t}{\mathrm{d}x}=\dfrac{1}{\frac{\mathrm{d}x}{\mathrm{d}t}}=\dfrac{1}{1+\cos t}$<br>$\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\mathrm{d}y}{\mathrm{d}t}\times\dfrac{\mathrm{d}t}{\mathrm{d}x}$<br>$=\dfrac{\sin t}{1+\cos t}$<br>$=\dfrac{2\sin\frac{t}{2}\cos\frac{t}{2}}{2\cos^2\frac{t}{2}}$<br>$=\dfrac{\sin\frac{t}{2}}{\cos\frac{t}{2}}$<br>$=\tan\dfrac{t}{2}$ (shown) | |

# 9 Applications of Differentiation

**9.1** The curve $C$ has equation $x^3+y^3-3xy=k$, where $k$ is a positive constant.

$C$ intersects the $x$-axis at the point $A$.

**(a)** Find the equation of the tangent to $C$ at $A$. Give your answer in the form $y=px+q$, where $p$ and $q$ are constants to be found in terms of $k$. [5]

**(b)** There are two points on $C$ where the tangents are parallel to the $y$-axis. Given that $k=3$, find the exact $y$-coordinates of these two points. [3]

**[EJC 2025 Prelim P2 Q1]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[5]** | Differentiating $C$ w.r.t. $x$,<br>$3x^2+3y^2\frac{dy}{dx}-3x\frac{dy}{dx}-3y=0$<br>$\frac{dy}{dx}=\frac{3y-3x^2}{3y^2-3x}=\frac{y-x^2}{y^2-x}$<br>When $y=0$,<br>$x^3=k$,<br>$x=k^{\frac{1}{3}}$<br>$\frac{dy}{dx}=x=k^{\frac{1}{3}}$<br>$y=k^{\frac{1}{3}}\left(x-k^{\frac{1}{3}}\right)=k^{\frac{1}{3}}x-k^{\frac{2}{3}}$ | |
| **(b)**<br>**[3]** | For tangent to be parallel to $y$-axis, $\frac{dy}{dx}$ must be undefined.<br>Hence,<br>$y^2-x=0$<br>$x=y^2$<br>Substituting into $C$ and $k=3$,<br>$y^6+y^3-3y^3-k=0$<br>$y^6-2y^3-3=0$<br>Factorising,<br>$(y^3-3)(y^3+1)=0$<br>$y^3=3$ or $y^3=-1$<br>$y=3^{\frac{1}{3}}$ or $y=-1$ | |

**9.2** A curve $C$ has parametric equations $x=\sqrt{t}, y=t^2+4$, where $0 \leqslant t \leqslant 16$.

**(i)** Find the equation of the tangent to $C$ at the point $P$ with parameter $p$. [3]

**(ii)** Given that the tangent in part **(i)** passes through the point $(0, 1)$, find the exact coordinates of the point $P$. [3]

**[MI 2025 Prelim P2 Q2(i), (ii)]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[3]** | $x=\sqrt{t}, \quad y=t^2+4$<br>$\frac{\mathrm{d}x}{\mathrm{d}t}=\frac{1}{2\sqrt{t}}, \quad \frac{\mathrm{d}y}{\mathrm{d}t}=2t$<br>$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{2t}{\frac{1}{2\sqrt{t}}}=4t^{\frac{3}{2}}$<br>Note: $t=p$<br>The eqn of tangent to $C$ at $P\left(\sqrt{p}, p^2+4\right)$:<br>$y-(p^2+4)=4p^{\frac{3}{2}}\left(x-\sqrt{p}\right)$<br>$y=4p^{\frac{3}{2}}x-3p^2+4$ | |
| **(ii)** **[3]** | The eqn of tangent of $C$ at $P$ passes through $(0, 1)$<br>At $(0, 1)$:<br>$1=4p^{\frac{3}{2}}(0)-3p^2+4$<br>$3p^2=3$<br>$p=1 \quad$ (reject $-1 \because p \geqslant 0$)<br>When $p=1$<br>$x=1, \ y=5$<br>$P(1, 5)$ | |

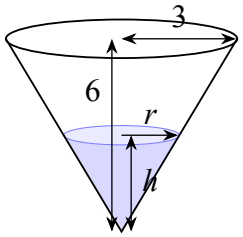
**9.3** A conical water tank is being filled with water at a constant rate of 0.2 $m^3s^{-1}$. The tank has a height of 6 m and a radius at the top of 3 m.

**(a)** Show that the volume of the water in the tank, $V$, can be expressed as $\frac{2}{3}\pi r^3$, where $r$ is the radius of the top water surface. [1]

**(b)** Find the exact rate of change of $r$ when the height of the water is 4 m. [3]

[The volume of a cone of base radius $r$ and height $h$ is given by $V=\frac{1}{3}\pi r^2 h$.]

**[RVHS 2025 Promo Q1]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)**<br>**[1]** |  <br><br>$V=\frac{1}{3}\pi r^2 h,$ where $h$ is the height of the water.<br>$=\frac{1}{3}\pi r^2(2r),$ since $\frac{h}{r}=\frac{6}{3}=\frac{2}{1}$ by similar triangles.<br>$=\frac{2}{3}\pi r^3$. (Shown) | |
| **(b)**<br>**[3]** | $\frac{dV}{dr}=2\pi r^2$<br>When $h=4$ (i.e. when $r=2$),<br>$\frac{dr}{dt}=\frac{dr}{dV}\times\frac{dV}{dt}$<br>$=\frac{1}{2\pi r^2}\times 0.2$<br>$=\frac{1}{8\pi}\times 0.2$<br>$=\frac{1}{40\pi}$ m $s^{-1}$ | |

**9.4** [The arc length $L$ of a sector of radius $a$ and angle $\theta$ is given by $L = a\theta$;
The volume $V$ of a cone of base radius $r$ and height $h$ is given by $V = \frac{1}{3}\pi r^2 h$ ]

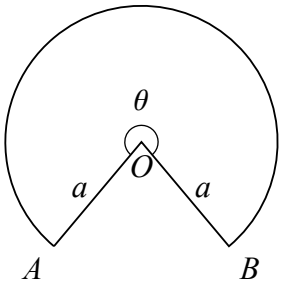


Figure 1

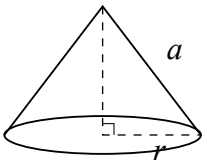


Figure 2

A metal sheet is shaped as a circular sector with angle $\theta$ and fixed radius $a$, as shown in Figure 1. This sector is then formed into a right circular cone with slant height $a$ by joining the two radii $OA$ and $OB$ together as shown in Figure 2.

**(a)** If $\theta$ is measured in radians, state the base radius of the cone, $r$, in term of $a$ and $\theta$.

[1]

**(b)** Show that the volume of the cone, $V$, is given by

$$V^2 = \frac{a^6}{576\pi^4}\theta^4 (4\pi^2 - \theta^2) .$$

[3]

**(c)** Using differentiation, find the exact value of $\theta$ that will maximise $V$. [4]

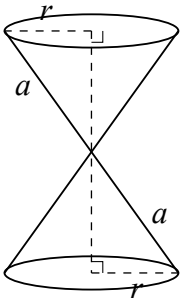


Figure 3

Two identical cones, as described in part **(c)**, are joined together tip-to-tip to form a new object, as shown in Figure 3. The upper cone is initially filled with water, which leaks out into the lower cone at a rate of $\frac{\pi}{15}$ cm$^3$ s$^{-1}$. At time $t$ seconds, the height of the water in the upper cone is $y$.

**(d)** Show that the volume of water in the cone, $W$ cm$^3$, is given by $W = \frac{2\pi y^3}{3}$. [2]

**(e)** Calculate the rate of change of $y$ when $y=2$. [2]

**[TJC 2025 Prelim P2 Q4]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[1]** | Circumference of base = length of arc AB<br>$2\pi r = a\theta$<br>$r = \frac{a}{2\pi}\theta$ | |
| **(b)** **[3]** | (i) Let $h$ = height of cone<br>$h^2 = a^2 - r^2$<br>$= a^2 - \frac{a^2}{4\pi^2}\theta^2$<br>$= \frac{a^2}{4\pi^2}(4\pi^2 - \theta^2)$<br>$V = \frac{1}{3}\pi r^2 h$<br>$V^2 = \frac{1}{9}\pi^2 r^4 h^2$<br>$= \frac{1}{9}\pi^2 \left(\frac{a^4}{16\pi^4}\right) \cdot \frac{a^2}{4\pi^2}(4\pi^2 - \theta^2)$<br>$= \frac{a^6}{576\pi^4}\theta^4(4\pi^2 - \theta^2)$ | |

**(c)**
**[4]**

**Method 1**

$$\frac{\mathrm{d}V^2}{\mathrm{d}\theta} = \frac{a^6}{576\pi^4}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(4\pi^2\theta^4 - \theta^6\right)$$
$$= \frac{a^6}{576\pi^4}\left(16\pi^2\theta^3 - 6\theta^5\right)$$
$$= \frac{a^6}{576\pi^4}\cdot 2\theta^3\left(8\pi^2 - 3\theta^2\right)$$

$\frac{\mathrm{d}V^2}{\mathrm{d}\theta} = 0$ when $\theta^3\left(8\pi^2 - 3\theta^2\right) = 0$

$$\theta^2 = \frac{8\pi^2}{3} \quad \Rightarrow \quad \theta = 2\pi\sqrt{\frac{2}{3}}$$

$$\frac{\mathrm{d}^2V^2}{\mathrm{d}\theta^2} = \frac{a^6}{576\pi^4}\left(48\pi^2\theta^2 - 30\theta^4\right)$$

When $\theta = 2\pi\sqrt{\frac{2}{3}}$,

$$\frac{\mathrm{d}^2V^2}{\mathrm{d}\theta^2} = \frac{a^6}{576\pi^4}\left(48\pi^2\left(\frac{8\pi^2}{3}\right) - 30\left(\frac{64\pi^4}{9}\right)\right) = -\frac{4a^6}{27} < 0$$

Therefore $V$ is max when $\theta = 2\pi\sqrt{\frac{2}{3}}$

**Method 2**

$$V^2 = \frac{a^6}{576\pi^4}\theta^4\left(4\pi^2 - \theta^2\right)$$

Differentiating w.r.t. $\theta$,

$$2V\frac{\mathrm{d}V}{\mathrm{d}\theta} = \frac{a^6}{576\pi^4}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(4\pi^2\theta^4 - \theta^6\right)$$
$$= \frac{a^6}{576\pi^4}\left(16\pi^2\theta^3 - 6\theta^5\right)$$
$$= \frac{a^6}{576\pi^4}\cdot 2\theta^3\left(8\pi^2 - 3\theta^2\right)$$

$\frac{\mathrm{d}V}{\mathrm{d}\theta} = 0$ when $\theta^3\left(8\pi^2 - 3\theta^2\right) = 0$

$$\theta^2 = \frac{8\pi^2}{3} \quad \Rightarrow \quad \theta = 2\pi\sqrt{\frac{2}{3}} \quad (\because \theta > 0)$$

$$2V\frac{\mathrm{d}^2V}{\mathrm{d}\theta^2} + 2\left(\frac{\mathrm{d}V}{\mathrm{d}\theta}\right)^2 = \frac{a^6}{576\pi^4}\left(48\pi^2\theta^2 - 30\theta^4\right)$$

When $\theta = 2\pi\sqrt{\frac{2}{3}}$, $\frac{\mathrm{d}V}{\mathrm{d}\theta} = 0$

$$2V\frac{\mathrm{d}^2V}{\mathrm{d}\theta^2} = \frac{a^6}{576\pi^4}\left(48\pi^2\left(\frac{8\pi^2}{3}\right) - 30\left(\frac{64\pi^4}{9}\right)\right) = -\frac{4a^6}{27} < 0$$

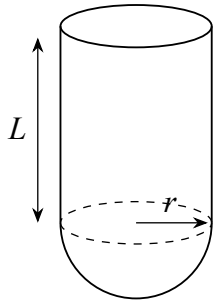
Since $V > 0$, $\frac{\mathrm{d}^2V}{\mathrm{d}\theta^2} < 0$

Therefore $V$ is max when $\theta = 2\pi\sqrt{\frac{2}{3}}$

| | | |
|---|---|---|
| **(d)**<br>**[2]**<br><br><br><br><br><br><br><br>**(e)**<br>**[2]** | Note that: $\theta = 2\pi\sqrt{\frac{2}{3}}$<br>$r = \frac{a}{2\pi}\theta = \frac{a}{2\pi}\left(2\pi\sqrt{\frac{2}{3}}\right) = a\sqrt{\frac{2}{3}}$<br>$h^2 = a^2 - r^2 = a^2 - \left(a\sqrt{\frac{2}{3}}\right)^2 = \frac{1}{3}a^2 \quad \Rightarrow h = \frac{1}{\sqrt{3}}a$<br>$\frac{r}{h} = a\sqrt{\frac{2}{3}} \times \frac{\sqrt{3}}{a} = \sqrt{2} \quad \Rightarrow r = \sqrt{2}h$<br>$W = \frac{1}{3}\pi r_w^2 h_w = \frac{1}{3}\pi\left(\sqrt{2}y\right)^2 y = \frac{2\pi y^3}{3}$<br>$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{\mathrm{d}W}{\mathrm{d}y} \times \frac{\mathrm{d}y}{\mathrm{d}t}$<br>$-\frac{\pi}{15} = 2\pi y^2 \times \frac{\mathrm{d}y}{\mathrm{d}t}$<br>$\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{1}{30y^2}$<br>When $y = 2$, $\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{1}{120}$ cm s$^{-1}$<br>Rate of change of $y$ is $-\frac{1}{120}$ cm s$^{-1}$. | |

**9.5** [The surface area and volume of a sphere are given by $4\pi r^2$ and $\frac{4}{3}\pi r^3$ respectively.]

An aquarist who has a fixed budget of \$$k$ wants to make a goldfish sanctuary. His design of the tank consists of a circular cover, a cylinder of length $L$ cm and a hemispherical base of radius $r$ cm neatly fitted together. The costs of the material per cm$^2$ used for the cover, the cylindrical surface and hemispherical base are \$20, \$10 and \$30 respectively. Assume that the tank is made of material with negligible thickness.

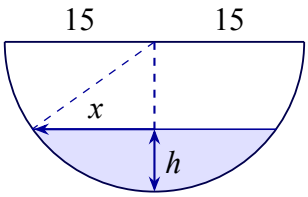



**(a)** Show that the volume of the tank is $V$ cm$^3$, where $V=\frac{k}{20}r-\frac{10}{3}\pi r^3$. [3]

**(b)** As $r$ varies, find the cost of the material used to make the hemispherical base in terms of $k$ when $V$ is at its maximum value. You need to show that $V$ is a maximum. [3]

The aquarist has fixed the radius of the cylinder to be 15 cm. Initially, there is some water in the hemispherical part of the tank. However, due to a defect in the tank, water is leaking at a constant rate of 20 cm$^3$ per minute.

The depth of the water at the hemispherical bottom is denoted by $h$ cm, and the radius of the water surface is $x$ cm. The volume of water in the hemispherical bottom is given by $W$ cm$^3$, where $W=\frac{\pi h\,(3x^2+h^2)}{6}$.




**(c)** By formulating a relationship between $x$ and $h$, find, when $h=3$, the rate of change of

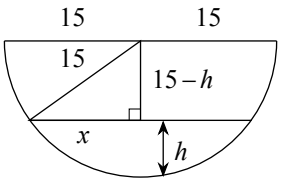
**(i)** the depth of water, [3]

**(ii)** the radius of the water surface. [3]

**[DHS 2025 Prelim P1 Q11]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $k=2\pi r^2(30)+2\pi rL(10)+\pi r^2(20)$<br>$k=80\pi r^2+20\pi rL$<br>$k=20\pi r(4r+L)$<br>$\Rightarrow L=\dfrac{k}{20\pi r}-4r$<br>$V=\dfrac{2}{3}\pi r^3+\pi r^2L$<br>$=\dfrac{2}{3}\pi r^3+\pi r^2\left(\dfrac{k}{20\pi r}-4r\right)$<br>$=\dfrac{2}{3}\pi r^3+\dfrac{kr}{20}-4\pi r^3$<br>$=\dfrac{k}{20}r-\dfrac{10}{3}\pi r^3$ (shown) | |
| **(b)** **[3]** | For maximum $V$ as $r$ varies,<br>$\dfrac{\mathrm{d}V}{\mathrm{d}r}=0$<br>$\dfrac{\mathrm{d}V}{\mathrm{d}r}=\dfrac{k}{20}-10\pi r^2$<br>$r=\sqrt{\dfrac{k}{200\pi}}$ (reject $r<0$)<br>$\dfrac{\mathrm{d}^2V}{\mathrm{d}r^2}=-20\pi r<0$<br>Therefore, $V$ is maximum.<br>Cost of material for the hemispherical base<br>$=2\pi r^2(30)=2\pi\left(\sqrt{\dfrac{k}{200\pi}}\right)^2(30)=\dfrac{3k}{10}$ dollars | |

**(c)(i)**
**[3]**




By Pythagoras' Theorem,

$$x^2 = 15^2 - (15-h)^2$$

$$x^2 = (15-(15-h))\,(15+(15-h))$$

$$x^2 = h(30-h)$$

$$W = \frac{\pi h\,(3x^2+h^2)}{6} = \frac{\pi h\,(3\,(h(30-h))+h^2)}{6}$$

$$W = \frac{\pi h\,(90h-2h^2)}{6} = \frac{\pi\,(45h^2-h^3)}{3}$$

$$\frac{\mathrm{d}W}{\mathrm{d}h} = \frac{\pi\,(90h-3h^2)}{3}$$

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{\mathrm{d}W}{\mathrm{d}h} \times \frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\pi\,(90h-3h^2)}{3} \times \frac{\mathrm{d}h}{\mathrm{d}t}$$

At $h = 3$,

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\frac{\mathrm{d}W}{\mathrm{d}t}}{\frac{\mathrm{d}W}{\mathrm{d}h}} = \frac{-20}{\frac{\pi(90(3)-3(3)^2)}{3}}$$

$$= -0.078595 = -0.0786 \text{ cm/min (3 s.f.)}$$

**(c)(ii)**
**[3]**

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}h} \times \frac{\mathrm{d}h}{\mathrm{d}t}$$

$$x^2 = h(30-h) \Rightarrow x = \sqrt{h(30-h)} = \sqrt{30h-h^2} \quad (x>0)$$

$$\frac{\mathrm{d}x}{\mathrm{d}h} = \frac{30-2h}{2\sqrt{30h-h^2}} = \frac{15-h}{\sqrt{30h-h^2}}$$

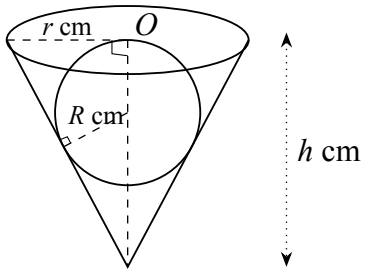
At $h = 3$,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \left(\frac{15-3}{\sqrt{30(3)-(3)^2}}\right) \times (-0.078595)$$

$$= -0.10479 = -0.105 \text{ cm/min (3sf)}$$

**9.6** [It is given that the volume of cone with radius $r$ and height $h$ is $\frac{1}{3}\pi r^2 h$ ]

A water filter system takes the shape of an inverted right circular cone with a circular top of radius $r$ cm and height $h$ cm. It contains a spherical filter which the water passes through during filtration. The sphere has a constant radius $R$ cm and is inscribed in the cone such that the sphere is in contact with the center of the circular top, $O$, and the slant surface of the cone as shown in the diagram below.

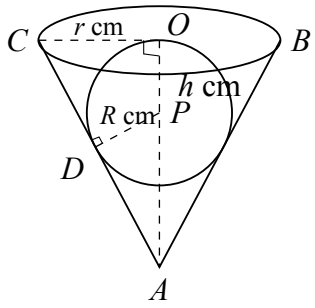



**(a)** Show that $r^2 = \dfrac{h^2R^2}{h^2 - 2hR}$. [3]

**(b)** As $h$ varies, find the minimum volume of the cone in terms of $R$. [5]

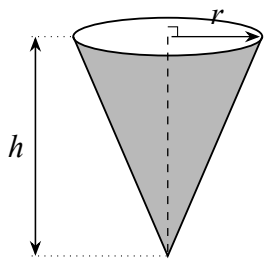
[You do not need to show that the volume is minimum]

**[TJC 2024 Promo Q12]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[3]** |  <br>Let centre of the sphere be $P$.<br>Using similar triangles $APD$ and $ACO$,<br>$\frac{h-R}{AC}=\frac{R}{r}$<br>$AC=\frac{rh-rR}{R}$<br>By Pythagoras' Theorem, $AC^2=h^2+r^2$<br>$\left(\frac{rh-rR}{R}\right)^2=h^2+r^2$<br>$r^2h^2-2r^2hR+r^2R^2=h^2R^2+r^2R^2$<br>$r^2\left(h^2-2hR\right)=h^2R^2$<br>$\therefore r^2=\frac{h^2R^2}{h^2-2hR}$<br>**OR**<br>Let centre of the sphere be $P$.<br>$AD=\sqrt{(AP^2-PD^2)}$<br>$=\sqrt{((h-R)^2-R^2)}$<br>$=\sqrt{(h^2-2hR)}$<br>Using congruent triangles $PCO$ and $PCD$, $CO=CD=r$ cm.<br>$AC=AD+DC$<br>$=\sqrt{(h^2-2hR)}+r$<br>Since by Pythagoras' theorem, $AC^2=h^2+r^2$<br>$h^2+r^2=h^2-2hR+2r\sqrt{(h^2-2hR)}+r^2$<br>$\therefore r^2=\frac{h^2R^2}{h^2-2hR}$ | |

| | | |
|---|---|---|
| **(b)**<br>**[5]** | Volume of cone, $V$<br>$= \frac{1}{3}\pi r^2 h$<br>$= \frac{1}{3}\pi \frac{h^2R^2}{h^2-2hR}h$<br>$= \frac{1}{3}\pi R^2 \frac{h^3}{h^2-2hR}$<br>$= \frac{1}{3}\pi R^2 \frac{h^2}{h-2R}$<br>$\frac{dV}{dh} = \frac{1}{3}\pi R^2 \left[\frac{2h(h-2R)-h^2}{(h-2R)^2}\right]$<br>$= \frac{1}{3}\pi R^2 \left[\frac{h^2-4hR}{(h-2R)^2}\right]$<br>$= \frac{1}{3}\pi R^2 \frac{h(h-4R)}{(h-2R)^2}$<br>$\frac{dV}{dh} = 0$<br>$\frac{1}{3}\pi R^2 \frac{h(h-4R)}{(h-2R)^2} = 0$<br>$h = 4R$ or $h = 0$ (reject $\because h > 0$)<br>$\therefore$ minimum $V = \frac{1}{3}\pi R^2 \frac{(4R)^2}{4R-2R} = \frac{8}{3}\pi R^3$ cm$^3$ | |

**9.7**




A cone-shaped cup is made of paper of negligible thickness to hold $20\pi$ cm$^3$ of liquid. The open inverted cone has radius $r$ cm and height $h$ cm as shown in the diagram above.

The external surface area of the cup is denoted by $A$ cm$^2$. The manufacturer wants to reduce the cost of production by minimizing the value of $A$.

**(a)** Find $h$ in terms of $r$. [1]

**(b)** By considering $A^2$ or otherwise, show that $A\dfrac{\mathrm{d}A}{\mathrm{d}r} = \pi^2\left(2r^3 - \dfrac{3600}{r^3}\right)$. [3]

**(c)** Find the exact value of $r$ that gives the minimum value of $A$, proving that $A$ is a minimum. Find also the ratio of the radius to the height, $\dfrac{r}{h}$, giving your answer in terms of $k\sqrt{2}$, where $k$ is a constant to be determined. [5]

The manufacturer decides to make paper cups at minimum external surface area using the ratio $\dfrac{r}{h}$ found in part **(c)**.

**(d)** The cup is being filled completely with water. However, there is a small hole at the bottom of the cup that causes water to leak out at the rate of 3 cm$^3$ per second. Find the rate of decrease of the depth of the water at the instant when the depth is 2 cm. [3]

[It is given that the volume of a circular cone with base radius $r$ and height $h$ is $\dfrac{1}{3}\pi r^2 h$ and the curved surface area is $\pi rl$, where $l$ is the slant height of the cone.]

**[CJC 2024 Promo Q10]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)**<br>**[1]** | Volume of cone $= \dfrac{1}{3}\pi r^2 h = 20\pi$<br>$\dfrac{r^2 h}{3} = 20$<br>$h = \dfrac{60}{r^2}$ | |

**(b)**
**[3]**

**Method 1: Considering $A^2$**

$$A = \pi rl$$
$$= \pi r\sqrt{r^2+h^2}$$
$$= \pi r\sqrt{r^2+\left(\frac{60}{r^2}\right)^2}$$
$$= \pi r\sqrt{r^2+\frac{3600}{r^4}}$$
$$A^2 = \pi^2 r^2\left(r^2+\frac{3600}{r^4}\right)$$
$$= \pi^2\left(r^4+\frac{3600}{r^2}\right)$$
$$= \pi^2\left(r^4+3600r^{-2}\right)$$

Differentiating implicitly with respect to $r$:

$$2A\frac{\mathrm{d}A}{\mathrm{d}r} = \pi^2\left(4r^3-7200r^{-3}\right)$$
$$A\frac{\mathrm{d}A}{\mathrm{d}r} = \pi^2\left(2r^3-\frac{3600}{r^3}\right) \quad \text{(shown)}$$

**Method 2: "Otherwise" Method**

$$A = \pi rl$$
$$= \pi r\sqrt{r^2+h^2}$$
$$= \pi r\sqrt{r^2+\left(\frac{60}{r^2}\right)^2}$$
$$= \pi r\sqrt{r^2+\frac{3600}{r^4}}$$
$$= \pi r\left(r^2+3600r^{-4}\right)^{\frac{1}{2}}$$

Differentiating with respect to $r$:

$$\frac{\mathrm{d}A}{\mathrm{d}r} = \pi\left[r\cdot\frac{1}{2}\left(r^2+3600r^{-4}\right)^{-\frac{1}{2}}\left(2r-4\cdot3600r^{-5}\right)+\left(r^2+3600r^{-4}\right)^{\frac{1}{2}}\right]$$
$$= \pi\left(r^2+3600r^{-4}\right)^{-\frac{1}{2}}\left[r^2-7200r^{-4}+r^2+3600r^{-4}\right]$$
$$= \pi\left(r^2+3600r^{-4}\right)^{-\frac{1}{2}}\left[2r^2-3600r^{-4}\right]$$
$$= \frac{\pi}{\sqrt{r^2+3600r^{-4}}}\left(2r^2-\frac{3600}{r^4}\right)$$
$$A\frac{\mathrm{d}A}{\mathrm{d}r} = \pi r\sqrt{r^2+3600r^{-4}}\cdot\frac{\pi}{\sqrt{r^2+3600r^{-4}}}\left(2r^2-\frac{3600}{r^4}\right)$$
$$= \pi^2 r\left(2r^2-\frac{3600}{r^4}\right)$$
$$= \pi^2\left(2r^3-\frac{3600}{r^3}\right) \quad \text{(shown)}$$

**(c)**
**[5]**

At stationary point, $\frac{\mathrm{d}A}{\mathrm{d}r} = 0$

$$\pi^2\left(2r^3-\frac{3600}{r^3}\right)=0$$

$$2r^3-\frac{3600}{r^3}=0$$

$$2r^3=\frac{3600}{r^3}$$

$$r^6=1800$$

$$r=(1800)^{\frac{1}{6}}$$

To prove minimum:

**Method 1: Using 1st derivative test**

| $r$ | $\left((1800)^{\frac{1}{6}}\right)^-$ | $(1800)^{\frac{1}{6}}$ | $\left((1800)^{\frac{1}{6}}\right)^+$ |
|---|---|---|---|
| Sign of $\frac{\mathrm{d}A}{\mathrm{d}r}$ | $<0$ | 0 | $>0$ |
| Tangent | ＼ | — | ／ |

$r=(1800)^{\frac{1}{6}}$ gives the minimum $A$.

**Method 2: Using 2nd derivative test**

From the result $A\frac{\mathrm{d}A}{\mathrm{d}r}=\pi^2\left(2r^3-\frac{3600}{r^3}\right)$, using implicit differentiation to obtain:

$$A\frac{\mathrm{d}^2A}{\mathrm{d}r^2}+\left(\frac{\mathrm{d}A}{\mathrm{d}r}\right)^2=\pi^2\left(6r^2+\frac{10800}{r^4}\right)$$

At $\frac{\mathrm{d}A}{\mathrm{d}r}=0$ and $r=(1800)^{\frac{1}{6}}$, $\frac{\mathrm{d}^2A}{\mathrm{d}r^2}=\frac{\pi^2}{A}\left(6r^2+\frac{10800}{r^4}\right)>0$ (minimum)

$r=(1800)^{\frac{1}{6}}$ gives the minimum $A$.

**Method 3:**

$$\frac{r}{h}=\frac{r}{\left(\frac{60}{r^2}\right)}$$

$$=\frac{r^3}{60}$$

$$=\frac{\left(1800^{\frac{1}{6}}\right)^3}{60}$$

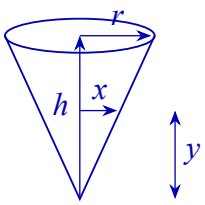
$$=\frac{\sqrt{1800}}{60}$$

$$=\frac{30\sqrt{2}}{60}$$

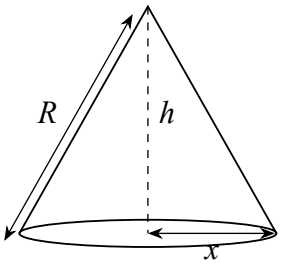
$$=\frac{\sqrt{2}}{2},\ \text{where } k=\frac{1}{2}$$

**Method 4:**

| | | |
|---|---|---|
| | $h = \frac{60}{r^2}$<br>$= \frac{60}{\left(1800^{\frac{1}{6}}\right)^2}$<br>$= \frac{60}{1800^{\frac{1}{3}}}$<br>$\frac{r}{h} = \frac{1800^{\frac{1}{6}}}{\left(\frac{60}{1800^{\frac{1}{3}}}\right)}$<br>$= \frac{\sqrt{1800}}{60}$<br>$= \frac{30\sqrt{2}}{60}$<br>$= \frac{\sqrt{2}}{2}$, where $k = \frac{1}{2}$ | |

| | | |
|---|---|---|
| **(d)**<br>**[3]** | Let $V$ be the volume of water in the cone when radius of water surface is $x$ cm and depth is $y$ cm.<br>By similar triangles,<br>  <br><br>$\frac{x}{y}=\frac{r}{h}=\frac{\sqrt{2}}{2}$<br>$x=\frac{\sqrt{2}}{2}y$<br>$V=\frac{1}{3}\pi x^2 y$<br>$=\frac{1}{3}\pi\left(\frac{\sqrt{2}}{2}y\right)^2 y$<br>$=\frac{1}{6}\pi y^3$<br>$\frac{dV}{dy}=\frac{1}{2}\pi y^2$<br>$\frac{dV}{dt}=\frac{dV}{dy}\times\frac{dy}{dt}$<br>$=\frac{1}{2}\pi y^2\times\frac{dy}{dt}$<br>When $y=2$,<br>$-3=\frac{1}{2}\pi(2)^2\times\frac{dy}{dt}$<br>$\frac{dy}{dt}=-\frac{3}{2\pi}$<br>Rate of decrease of depth of water is $\frac{3}{2\pi}$ cm/s or 0.477 cm/s (3 s.f.) | |

**9.8**




The diagram shows a right cone with a slant edge of fixed length $R$ cm, height $h$ cm and radius $x$ cm.

Show that $9V^2 = \pi^2 x^4 (R^2 - x^2)$, where $V$ is the volume of the cone.

Show further that, as $x$ varies, $V$ is a maximum when $x^2 = \frac{2}{3}R^2$. [7]

**[EJC 2018 Promo Q7]**

| Solution | | Comments |
|---|---|---|
| **[7]** | $h = \sqrt{R^2 - x^2}$<br>$V = \frac{1}{3}\pi x^2 h$<br>$= \frac{1}{3}\pi x^2 \sqrt{R^2 - x^2}$<br>$V^2 = \frac{1}{9}\pi^2 x^4 \left(\sqrt{R^2 - x^2}\right)^2$<br>$9V^2 = \pi^2 x^4 (R^2 - x^2)$ (shown)<br>Differentiate w.r.t. $x$,<br>$9V^2 = \pi^2 R^2 x^4 - \pi^2 x^6$<br>$18V\frac{dV}{dx} = 4\pi^2 R^2 x^3 - 6\pi^2 x^5$ — (*)<br>When the volume is maximum, $\frac{dV}{dx} = 0$<br>$-6\pi^2 x^5 + 4\pi^2 R^2 x^3 = 0$<br>$2\pi^2 x^3 (-3x^2 + 2R^2) = 0$<br>$-3x^2 + 2R^2 = 0$ or $x = 0$ (reject $\because x > 0$)<br>$x^2 = \frac{2}{3}R^2 \Rightarrow R^2 = \frac{3}{2}x^2$<br>**Check that $V$ is indeed a maximum:**<br><u>Second derivative method</u><br>Differentiating (*) wrt $x$<br>$18V\frac{d^2V}{dx^2} + 18\left(\frac{dV}{dx}\right)^2 = 12\pi^2 R^2 x^2 - 30\pi^2 x^4$<br>Substitute $x^2 = \frac{2}{3}R^2$, $\frac{dV}{dx} = 0$, | |

$$18V\frac{\mathrm{d}^2V}{\mathrm{d}x^2}+18(0)^2=12\pi^2R^2\left(\frac{2R^2}{3}\right)-30\pi^2\left(\frac{2R^2}{3}\right)^2$$

$$18V\frac{\mathrm{d}^2V}{\mathrm{d}x^2}=\frac{-40}{3}\pi^2R^4+8\pi^2R^4$$

$$18V\frac{\mathrm{d}^2V}{\mathrm{d}x^2}=-\frac{16}{3}\pi^2R^4$$

$$\frac{\mathrm{d}^2V}{\mathrm{d}x^2}=-\frac{16\pi^2R^4}{54V}<0$$

Therefore, $V$ is a maximum when $x^2=\frac{2}{3}R^2$.

<u>First derivative method</u>

From (*),

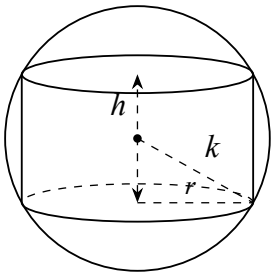
$$\frac{\mathrm{d}V}{\mathrm{d}x}=\frac{4\pi^2R^2x^3-6\pi^2x^5}{18V}=\frac{\pi^2x^3\,(2R^2-3x^2)}{9V}$$

(*Note to students*: Since $\frac{\pi^2x^3}{9V}>0$ for all $x$, we just need to consider $(2R^2-3x^2)$.)

| $x$ | $\left(\sqrt{\frac{2}{3}}R\right)^-$ | $\sqrt{\frac{2}{3}}R$ | $\left(\sqrt{\frac{2}{3}}R\right)^+$ |
|---|---|---|---|
| $\frac{\mathrm{d}V}{\mathrm{d}x}$ | $2R^2-3\left[\left(\sqrt{\frac{2}{3}}R\right)^-\right]^2$ $=2R^2-(2R^2)^->0$ | 0 | $2R^2-3\left[\left(\sqrt{\frac{2}{3}}R\right)^+\right]^2$ $=2R^2-(2R^2)^+<0$ |
| Slope | / | – | \ |

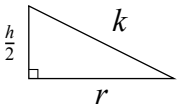
Therefore, $V$ is a maximum when $x^2=\frac{2}{3}R^2$.

**9.9**




A cylinder with height $h$ cm and base radius $r$ cm is inscribed within a sphere with fixed radius $k$ cm. The circumference of the circular top and bottom of the cylinder is in contact with the sphere (see figure above). By using differentiation, find, in terms of $k$, the exact value of $h$ for which the volume of the cylinder is maximum. [6]

**[TMJC 2024 Prelim P1 Q2]**

| **Solution** | | **Comments** |
|---|---|---|
| **[6]** | Let $V$ be the volume of the cylinder (in $\text{cm}^3$).<br>  <br><br>By Pythagoras Theorem,<br>$\left(\frac{h}{2}\right)^2 + r^2 = k^2$<br>$r^2 = k^2 - \frac{h^2}{4} \quad \text{— (1)}$<br>$V = \pi r^2 h$<br>$= \pi\left(k^2 - \frac{h^2}{4}\right)h \quad$ ,from (1)<br>$= \pi k^2 h - \frac{\pi}{4}h^3$<br>Differentiate w.r.t $h$,<br>$\frac{\mathrm{d}V}{\mathrm{d}h} = \pi k^2 - \frac{3\pi}{4}h^2$<br>For maximum $V$, $\frac{\mathrm{d}V}{\mathrm{d}h} = 0$<br>$\therefore \frac{\mathrm{d}V}{\mathrm{d}h} = \pi k^2 - \frac{3\pi}{4}h^2 = 0$<br>$k^2 - \frac{3}{4}h^2 = 0$<br>$h^2 = \frac{4k^2}{3}$<br>$h = \frac{2k}{\sqrt{3}} \quad (\because h > 0)$<br>**Testing $V$ is maximum**<br>**Method 1:** 1st derivative test | |

| $h$ | $\left(\frac{2k}{\sqrt{3}}\right)^-$ | $\frac{2k}{\sqrt{3}}$ | $\left(\frac{2k}{\sqrt{3}}\right)^+$ |
|---|---|---|---|
| $\frac{\mathrm{d}V}{\mathrm{d}h}$ | + | 0 | − |
| Slope | ↗ | — | ↘ |

$\therefore V$ is maximum when $h=\frac{2k}{\sqrt{3}}$.

**Method 2:** 2nd derivative test

Differentiate w.r.t $h$,

$\frac{\mathrm{d}^2V}{\mathrm{d}h^2}=-\frac{3\pi}{2}h$

When $h=\frac{2k}{\sqrt{3}}$, $\frac{\mathrm{d}^2V}{\mathrm{d}h^2}=-\frac{3\pi}{2}\left(\frac{2k}{\sqrt{3}}\right)=-\sqrt{3}\pi k<0 \quad (\because k>0)$

$\therefore V$ is maximum when $h=\frac{2k}{\sqrt{3}}$.

# 10 Power Series

**10.1** It is given that $y = \ln \sqrt{1+e^{3x}}$.

**(a)** Show that $2e^{2y}\frac{dy}{dx} = 3e^{3x}$. [2]

**(b)** By further differentiation of the result in part **(a)**, find the Maclaurin series for $y$, up to and including the term in $x^2$, giving the coefficients in exact form. [4]

**(c)** Deduce the Maclaurin series for $\ln\left(\frac{\sqrt{1+e^{-3x}}}{2}\right)$, up to and including the term in $x^2$. [2]

**[TJC 2025 Prelim P2 Q1]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | $y=\frac{1}{2}\ln(1+e^{3x})$<br>$e^{2y}=1+e^{3x}$<br>Diff. wrt $x$,<br>$2e^{2y}\frac{dy}{dx}=3e^{3x}$ (Shown) $-(1)$<br>Alternative solution:<br>$y=\ln\sqrt{1+e^{3x}}$<br>$\frac{dy}{dx}=\frac{3e^{3x}(1+e^{3x})^{-\frac{1}{2}}}{2(1+e^{3x})^{\frac{1}{2}}}=\frac{3e^{3x}}{2(1+e^{3x})}$<br>$2(1+e^{3x})\frac{dy}{dx}=3e^{3x}$<br>$2e^{2y}\frac{dy}{dx}=3e^{3x}$<br>$\left(\because y=\ln\sqrt{1+e^{3x}}=\frac{1}{2}\ln(1+e^{3x})\right.$ i.e. $\left.2y=\ln(1+e^{3x})\Rightarrow e^{2y}=1+e^{3x}\right)$ | |
| **(b)** **[4]** | Diff. wrt $x$,<br>$2e^{2y}\frac{d^2y}{dx^2}+4e^{2y}\left(\frac{dy}{dx}\right)^2=9e^{3x}$ —- (2)<br>When $x=0, y=\frac{1}{2}\ln(1+1)=\frac{1}{2}\ln 2$, $e^{2y}=e^{\ln 2}=2$<br>Sub into (1): $\frac{dy}{dx}=\frac{3}{4}$<br>Sub into (2): $2(2)\frac{d^2y}{dx^2}+8\left(\frac{3}{4}\right)^2=9 \Rightarrow \frac{d^2y}{dx^2}=\frac{9}{8}$<br>$y=\frac{1}{2}\ln 2+\frac{3}{4}x+\frac{9}{16}x^2+\ldots$ | |
| **(c)** **[2]** | $\ln\left(\frac{\sqrt{1+e^{-3x}}}{2}\right)=\ln\left(\sqrt{1+e^{3(-x)}}\right)-\ln 2$<br>$=-\ln 2+\frac{1}{2}\ln 2+\frac{3}{4}(-x)+\frac{9}{16}(-x)^2+\ldots$<br>$=-\frac{1}{2}\ln 2-\frac{3}{4}x+\frac{9}{16}x^2+\ldots$ | |

**10.2** It is given that $y = e^{\sin^{-1}(3x)}$, where $-\frac{1}{3} < x < \frac{1}{3}$.

**(a)** Show that $(1-9x^2)\frac{d^2y}{dx^2} - 9x\frac{dy}{dx} - 9y = 0$. [3]

**(b)** By differentiating the above result, find the Maclaurin series for $y$, up to and including the term in $x^3$. [4]

**(c)** Using your result in part **(b)** and the standard series in the List of Formulae (MF27), obtain the Maclaurin series for $\sin^{-1}(3x)$, up to and including the term in $x^3$. [3]

**[YIJC 2026 Prelim P1 Q10]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | **EITHER** $y = e^{\sin^{-1}(3x)}$ $\frac{dy}{dx} = 3 \cdot \frac{1}{\sqrt{1-9x^2}} e^{\sin^{-1}(3x)}$ $= \frac{3y}{\sqrt{1-9x^2}}$ $\sqrt{1-9x^2}\frac{dy}{dx} = 3y$ **OR** $y = e^{\sin^{-1}(3x)}$ $\ln y = \sin^{-1}(3x)$ Differentiating w.r.t. $x$, $\frac{1}{y}\frac{dy}{dx} = \frac{1}{\sqrt{1-9x^2}} \cdot 3$ $\sqrt{1-9x^2}\frac{dy}{dx} = 3y$ **THEN** **EITHER** Differentiating with respect to $x$, $\sqrt{1-9x^2}\frac{d^2y}{dx^2} + \frac{1}{2}(1-9x^2)^{-\frac{1}{2}}(-18x)\frac{dy}{dx} = 3\frac{dy}{dx}$ Multiplying throughout by $\sqrt{1-9x^2}$, $(1-9x^2)\frac{d^2y}{dx^2} - 9x\frac{dy}{dx} = 3\sqrt{1-9x^2}\frac{dy}{dx}$ $= 3(3y)$ $(1-9x^2)\frac{d^2y}{dx^2} - 9x\frac{dy}{dx} - 9y = 0$ (shown) — (1) **OR** Squaring both sides, $(1-9x^2)\left(\frac{dy}{dx}\right)^2 = 9y^2$ Differentiating with respect to $x$, $(1-9x^2)\cdot 2\left(\frac{dy}{dx}\right)\frac{d^2y}{dx^2} - 18x\left(\frac{dy}{dx}\right)^2 = 18y\frac{dy}{dx}$ Dividing throughout by $2\frac{dy}{dx}$, $(1-9x^2)\frac{d^2y}{dx^2} - 9x\left(\frac{dy}{dx}\right) = 9y$ $(1-9x^2)\frac{d^2y}{dx^2} - 9x\frac{dy}{dx} - 9y = 0$ (shown) | |
| **(b)** **[4]** | Differentiating with respect to $x$, | |

| | | |
|---|---|---|
| **(c)**<br>**[3]** | $(1-9x^2)\dfrac{d^3y}{dx^3}-18x\dfrac{d^2y}{dx^2}-9x\dfrac{d^2y}{dx^2}-9\dfrac{dy}{dx}-9\dfrac{dy}{dx}=0$<br>$(1-9x^2)\dfrac{d^3y}{dx^3}-27x\dfrac{d^2y}{dx^2}-18\dfrac{dy}{dx}=0 \quad \text{— (2)}$<br>When $x=0, y=1.\ \dfrac{dy}{dx}=3$<br>From (1): $\dfrac{d^2y}{dx^2}-0-9=0\Rightarrow\dfrac{d^2y}{dx^2}=9$<br>From (2): $\dfrac{d^3y}{dx^3}-0-18(3)=0\Rightarrow\dfrac{d^3y}{dx^3}=54$<br>$\therefore y=1+3x+\dfrac{9}{2!}x^2+\dfrac{54}{3!}x^3+\ldots$<br>$=1+3x+\dfrac{9}{2}x^2+9x^3+\ldots$<br>$e^{\sin^{-1}(3x)}=1+3x+\dfrac{9}{2}x^2+9x^3+\ldots$<br>$\sin^{-1}(3x)\approx\ln\left(1+3x+\dfrac{9}{2}x^2+9x^3\right)$<br>$\approx\left(3x+\dfrac{9}{2}x^2+9x^3\right)-\dfrac{1}{2}\left(3x+\dfrac{9}{2}x^2\right)^2+\dfrac{1}{3}(3x)^3$<br>$\approx 3x+\dfrac{9}{2}x^2+9x^3-\dfrac{1}{2}(9x^2+27x^3)+\dfrac{1}{3}(27x^3)$<br>$=3x+\dfrac{9}{2}x^3$ | |

**10.3** It is given that $\mathrm{f}(x)=\dfrac{1}{\sqrt{1-4x^2}}$ where $-\dfrac{1}{2}<x<\dfrac{1}{2}$.

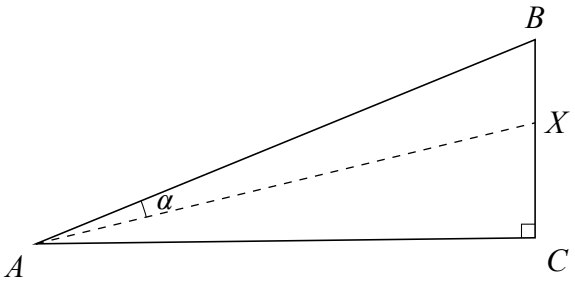
**(a)** Using standard series from the List of Formulae (MF27), find the Maclaurin's expansion of $\mathrm{f}(x)$, up to and including the term in $x^6$. [2]

**(b)** Hence find the first four non-zero terms of the Maclaurin series for $\sin^{-1}2x$. [4]

**[ACJC 2025 Prelim P2 Q2]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | $\mathrm{f}(x)=\dfrac{1}{\sqrt{1-4x^2}}=(1-4x^2)^{-\frac{1}{2}}$<br>$=1-\dfrac{1}{2}(-4x^2)+\dfrac{-\frac{1}{2}\left(-\frac{3}{2}\right)}{2!}(-4x^2)^2+\dfrac{-\frac{1}{2}\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}(-4x^2)^3+\ldots$<br>$=1+2x^2+6x^4+20x^6+\ldots$ | |
| **(b)** **[4]** | $\displaystyle\int\frac{1}{\sqrt{1-4x^2}}\,\mathrm{d}x=\int 1+2x^2+6x^4+20x^6\,\mathrm{d}x$<br>$\dfrac{1}{2}\sin^{-1}2x=x+\dfrac{2}{3}x^3+\dfrac{6}{5}x^5+\dfrac{20}{7}x^7+c$<br>$\sin^{-1}2x=2x+\dfrac{4}{3}x^3+\dfrac{12}{5}x^5+\dfrac{40}{7}x^7+d$<br>$\sin^{-1}0=0 \quad \therefore d=0$<br>$\sin^{-1}2x=2x+\dfrac{4}{3}x^3+\dfrac{12}{5}x^5+\dfrac{40}{7}x^7$ | |

**10.4** In the right-angled triangle $ABC$, angle $C$ is a right angle. $AB = 13$ cm, $BC = 5$ cm and $X$ is a point on $BC$ such that angle $BAX$ is $\alpha$ radians.




**(a)** Find the exact value of $\cos \angle BAC$. [1]

**(b)** By considering triangle $AXC$, or otherwise, show that $AX = \dfrac{156}{12\cos\alpha + 5\sin\alpha}$. [3]

**(c)** Given that $\alpha$ is a sufficiently small angle, show that

$$AX \approx 13 + p\alpha + q\alpha^2,$$

where $p$ and $q$ are constants to be determined exactly. [4]

**[EJC 2025 Prelim P1 Q3]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[1]** | $AC = \sqrt{13^2 - 5^2} = 12$<br>$\cos \angle BAC = \frac{AC}{AX} = \frac{12}{13}$ | |
| **(b)** **[3]** | $AX = \frac{12}{\cos(\angle BAC - \alpha)}$<br>$= \frac{12}{\cos \angle BAC \cos \alpha + \sin \angle BAC \sin \alpha}$<br>$= \frac{12}{\frac{12}{13}\cos \alpha + \frac{5}{13}\sin \alpha}$<br>$= \frac{156}{12\cos \alpha + 5\sin \alpha}$ (shown) | |
| **(c)** **[4]** | Using small angle approximation,<br>$AX \approx \frac{156}{12\left(1 - \frac{\alpha^2}{2}\right) + 5\alpha}$<br>$= 156\left[12\left(1 + \frac{5}{12}\alpha - \frac{\alpha^2}{2}\right)\right]^{-1}$<br>$= 156[12]^{-1}\left[1 + \left(\frac{5}{12}\alpha - \frac{\alpha^2}{2}\right)\right]^{-1}$<br>$= 13\left[1 - \left(\frac{5}{12}\alpha - \frac{\alpha^2}{2}\right) + \left(\frac{5}{12}\alpha - \frac{\alpha^2}{2}\right)^2 + \ldots\right]$<br>$= 13\left[1 - \frac{5}{12}\alpha + \frac{\alpha^2}{2} + \left(\frac{5}{12}\alpha\right)^2 + \ldots\right]$<br>$\approx 13 - \frac{65}{12}\alpha + \frac{1261}{144}\alpha^2$<br>$p = -\frac{65}{12}, \quad q = \frac{1261}{144}$ (shown) | |

**10.5** It is given that $\mathrm{f}(x) = \dfrac{1}{4+9x^2}$.

**(i)** Find $\int \mathrm{f}(x)\,\mathrm{d}x$. [2]

**(ii)** Find the binomial expansion for $\mathrm{f}(x)$, up to and including the term in $x^4$. Give the coefficients as exact fractions in their simplest form. [2]

**(iii)** Hence, find the Maclaurin series for $\tan^{-1}\dfrac{3x}{2}$. Give the coefficients as exact fractions in their simplest form. [3]

**(iv)** Use your series from part **(iii)** to estimate $\displaystyle\int_0^{0.5} \tan^{-1}\frac{3x}{2}\,\mathrm{d}x$, correct to 3 decimal places. [1]

**(v)** Use your calculator to find $\displaystyle\int_0^{0.5} \tan^{-1}\frac{3x}{2}\,\mathrm{d}x$, correct to 3 decimal places. [1]

**(vi)** Comparing your answers to parts **(iv)** and **(v)**, comment on the accuracy of your estimate in **(iv)** and how it can be improved. [2]

**[JPJC 2025 Prelim P1 Q9]**

| Solution | | Comments |
|---|---|---|
| **(i)** **[2]** | $\int \frac{1}{4+9x^2}\,\mathrm{d}x = \int \frac{1}{9\left(\frac{4}{9}+x^2\right)}\,\mathrm{d}x = \frac{1}{6}\tan^{-1}\frac{3x}{2}+K$ | |
| **(ii)** **[2]** | $\mathrm{f}(x) = (4+9x^2)^{-1} = \frac{1}{4}\left(1+\frac{9x^2}{4}\right)^{-1}$<br>$= \frac{1}{4}\left[1+(-1)\left(\frac{9x^2}{4}\right)+\frac{-1(-2)}{2}\left(\frac{9x^2}{4}\right)^2+\ldots\right]$<br>$= \frac{1}{4}\left(1-\frac{9x^2}{4}+\frac{81}{16}x^4+\ldots\right)$<br>$\approx \frac{1}{4}-\frac{9}{16}x^2+\frac{81}{64}x^4$ | |
| **(iii)** **[3]** | From **(i)** $\tan^{-1}\frac{3x}{2} = 6\int \frac{1}{4+9x^2}\,\mathrm{d}x + C, \quad$ where $C=-6K$<br>From **(ii)** $\tan^{-1}\frac{3x}{2} = 6\int\left(\frac{1}{4}-\frac{9}{16}x^2+\frac{81}{64}x^4\right)\mathrm{d}x + C$<br>$= 6\left(\frac{1}{4}x-\frac{3}{16}x^3+\frac{81}{320}x^5\right)+D$<br>$= \frac{3}{2}x-\frac{9}{8}x^3+\frac{243}{160}x^5+D$<br>$x=0,\ \tan^{-1}0=0 \Rightarrow D=0$<br>$\therefore\ \tan^{-1}\frac{3x}{2} = \frac{3}{2}x-\frac{9}{8}x^3+\frac{243}{160}x^5$ | |
| **(iv)** **[1]** | $\int_0^{0.5}\tan^{-1}\frac{3x}{2}\,\mathrm{d}x = \int_0^{0.5}\left(\frac{3}{2}x-\frac{9}{8}x^3+\frac{243}{160}x^5\right)\mathrm{d}x$<br>$= 0.174$ (to 3 dp) (Using GC) | |
| **(v)** **[1]** | From GC, $\int_0^{0.5}\tan^{-1}\frac{3x}{2}\,\mathrm{d}x = 0.173$ (to 3 dp) | |
| **(vi)** **[2]** | The estimate in **(iv)** is accurate up to 2 decimal places but not to 3 decimal places.<br>To improve the estimate, we can include higher-order terms in the Maclaurin series expansion of $\tan^{-1}\frac{3x}{2}$. | |

**10.6** It is given that $y = \ln(\cos x)$.

**(i)** Prove that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 1 = 0$. [2]

**(ii)** By differentiating further, find the first two non-zero terms of the expansion of $y$. [4]

**(iii)** Using the answer in **(ii)** and an appropriate expansion from MF26 up to and including the term in $x^4$, find an approximate value for $\int_0^{0.08} \ln\left(\frac{\cos^3 x}{1-x}\right) \mathrm{d}x$, leaving your answer correct to 7 decimal places. [2]

The calculated value of $\int_0^{0.08} \ln\left(\frac{\cos^3 x}{1-x}\right) \mathrm{d}x$ is given to be 0.0030328 correct to 7 decimal places.

**(iv)** Comparing the calculated value of $\int_0^{0.08} \ln\left(\frac{\cos^3 x}{1-x}\right) \mathrm{d}x$ given above to your answer in **(iii)**, and with reference to the value of $x$, comment on the accuracy of your approximation. [2]

**[SAJC 2024 Prelim P2 Q4]**

| Solution | | Comments |
|---|---|---|
| **(i)**<br>**[2]** | $y=\ln(\cos x)$ —- (1)<br>Therefore, $e^y=\cos x$<br>Differentiate with respect to $x$,<br>$e^y\left(\frac{dy}{dx}\right)=-\sin x$ —- (2)<br>Differentiate with respect to $x$,<br>$e^y\left(\frac{d^2y}{dx^2}\right)+\left(\frac{dy}{dx}\right)^2 e^y=-\cos x$<br>$e^y\left(\frac{d^2y}{dx^2}\right)+\left(\frac{dy}{dx}\right)^2 e^y=-e^y$<br>$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=-1$ —- (3)<br>$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2+1=0$ (shown)<br><u>Alternative Method</u><br>$y=\ln(\cos x)$<br>$\frac{dy}{dx}=\frac{1}{\cos x}(-\sin x)$<br>$=-\frac{\sin x}{\cos x}$<br>$\frac{d^2y}{dx^2}=-\frac{\cos x(\cos x)-\sin x(-\sin x)}{\cos^2 x}$<br>$=-\frac{\cos^2 x+\sin^2 x}{\cos^2 x}$<br>$=-1-\frac{\sin^2 x}{\cos^2 x}$<br>$=-1-\left(\frac{dy}{dx}\right)^2$<br>$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=-1$<br>$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2+1=0$ (shown) | |

**(ii)**
**[4]**

Differentiate with respect to $x$,

$$\frac{d^3y}{dx^3}+2\left(\frac{dy}{dx}\right)\left(\frac{d^2y}{dx^2}\right)=0 \quad \text{(shown)} \quad \text{—- (4)}$$

Differentiate with respect to $x$,

$$\frac{d^4y}{dx^4}+2\left[\frac{d^2y}{dx^2}\cdot\frac{d^2y}{dx^2}+\frac{d^3y}{dx^3}\cdot\frac{dy}{dx}\right]=0$$

$$\frac{d^4y}{dx^4}+2\left(\frac{dy}{dx}\right)\left(\frac{d^3y}{dx^3}\right)+2\left(\frac{d^2y}{dx^2}\right)^2=0 \quad \text{—— (5)}$$

Let $y=\ln(\cos x)=f(x)$

When $x=0$,

From (1), $y=f(0)=\ln(\cos 0)=0$

From (2), $\frac{dy}{dx}=f'(0)=0$

From (3), $\frac{d^2y}{dx^2}=f''(0)=-(0)^2-1=-1$

From (4), $\frac{d^3y}{dx^3}=f'''(0)=-2(0)\,x\,(-1)=0$

From (5), $\frac{d^4y}{dx^4}=f^4(0)=-2\left[(-1)^2+(0)-(0)\right]=-2$

Using Maclaurin's expansion,

$$f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\frac{f^4(0)}{4!}x^4+\ldots$$

$$=-\frac{1}{2}x^2-\frac{1}{12}x^4+\ldots$$

$$\ln(\cos x)=-\frac{1}{2}x^2-\frac{1}{12}x^4+\ldots$$

**(iii)**
**[2]**

$$\int_0^{0.08}\ln\left(\frac{\cos^3 x}{1-x}\right)dx$$

$$=\int_0^{0.08}\left[\ln(\cos x)^3-\ln(1-x)\right]dx$$

$$=\int_0^{0.08}\left[3\ln(\cos x)-\ln(1-x)\right]dx$$

$$\approx\int_0^{0.08}\left[3\left(-\frac{1}{2}x^2-\frac{1}{12}x^4\right)-\left(-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}\right)\right]dx$$

$$=0.0030327 \text{ (to 7 decimal places)}$$

**(iv)**
**[2]**

The two values are the same up to 6 decimal places, and the $x$ value is between 0 and 0.08, which is relatively small and close to 0. Hence, the approximation is relatively accurate for the integral.

# 11 Integration

**11.1 (a)** Find $\int \frac{5-4x}{\sqrt{1+2x-x^2}}\,\mathrm{d}x$. [3]

**[VJC 2025 Promo Q7(a)]**

| Solution | | Comments |
|---|---|---|
| **(a)**<br>**[3]** | $\int \frac{5-4x}{\sqrt{1+2x-x^2}}\,\mathrm{d}x$<br>$=\int 2\,(2-2x)\,(1+2x-x^2)^{-\frac{1}{2}}\,\mathrm{d}x+\int \frac{1}{\sqrt{1+2x-x^2}}\,\mathrm{d}x$<br>$=2\left(\frac{1}{\frac{1}{2}}\right)(1+2x-x^2)^{\frac{1}{2}}+\int \frac{1}{\sqrt{2-(x-1)^2}}\,\mathrm{d}x$<br>$=4\sqrt{1+2x-x^2}+\sin^{-1}\left(\frac{x-1}{\sqrt{2}}\right)+C$ | Spilt the numerator into $kf'(x)+m$ allows us to use standard form $\int f'(x)\,[f(x)]^n\,\mathrm{d}x$ and from MF 27 $\int \frac{1}{\sqrt{a^2-x^2}}\,\mathrm{d}x$.<br>Must write $+C$ |

**11.2 (b) (i)** Use the substitution $x=\tan\theta$ to show that

$$\int \frac{x^5}{(x^2+1)^4}\,\mathrm{d}x=\int \sin^m\theta\cos^n\theta\,\mathrm{d}\theta,$$

where $m$ and $n$ are integers to be determined. [2]

**(ii)** Hence, find the exact value of $\int_0^{\sqrt{3}} \frac{x^5}{(x^2+1)^4}\,\mathrm{d}x$. [3]

**[VJC 2025 Promo Q7(b)]**

| Solution | | Comments |
|---|---|---|
| **(b)(i)** **[2]** | $x=\tan\theta \quad\Rightarrow\quad \frac{\mathrm{d}x}{\mathrm{d}\theta}=\sec^2\theta$<br>$\int \frac{x^5}{(x^2+1)^4}\,\mathrm{d}x=\int \frac{\tan^5\theta}{(\tan^2\theta+1)^4}\sec^2\theta\,\mathrm{d}\theta$<br>$=\int \frac{\tan^5\theta}{\sec^8\theta}\sec^2\theta\,\mathrm{d}\theta$<br>$=\int \frac{\left(\frac{\sin^5\theta}{\cos^5\theta}\right)}{\left(\frac{1}{\cos^6\theta}\right)}\mathrm{d}\theta$<br>$=\int \sin^5\theta\cos\theta\,\mathrm{d}\theta$ | |
| **(b)(ii)** **[3]** | When $x=\sqrt{3}$, $\sqrt{3}=\tan\theta \Rightarrow \theta=\frac{\pi}{3}$<br>When $x=0$, $0=\tan\theta \Rightarrow \theta=0$<br>$\int_0^{\sqrt{3}} \frac{x^5}{(x^2+1)^4}\,\mathrm{d}x=\int_0^{\frac{\pi}{3}} \sin^5\theta\cos\theta\,\mathrm{d}\theta$<br>$=\frac{1}{6}\left[\sin^6\theta\right]_0^{\frac{\pi}{3}}$<br>$=\frac{1}{6}\left(\frac{\sqrt{3}}{2}\right)^6$<br>$=\frac{9}{128}$ | Sufficient working must be shown to demonstrate that calculator is not used here since question ask for "exact" |

**11.3 (c)** Find $\int e^{-x}\cos 3x\,dx$. Hence, find the exact value of $\int_0^{\pi} e^{-x}\cos 3x\,dx$. [5]

**[YIJC 2024 Prelim P1 Q9(c)]**

| Solution | | Comments |
|---|---|---|
| **(c)**<br>**[5]** | $\int e^{-x}\cos 3x\,dx$<br>$= -e^{-x}\cos 3x - \int -e^{-x}(-3\sin 3x)\,dx$<br>$= -e^{-x}\cos 3x - 3\int e^{-x}\sin 3x\,dx$<br>$= -e^{-x}\cos 3x - 3\left[-e^{-x}\sin 3x - \int -e^{-x}(3\cos 3x)\,dx\right]$<br>$= -e^{-x}\cos 3x + 3e^{-x}\sin 3x - 9\int e^{-x}\cos 3x\,dx$<br>$\int e^{-x}\cos 3x\,dx = -e^{-x}\cos 3x + 3e^{-x}\sin 3x - 9\int e^{-x}\cos 3x\,dx$<br>$10\int e^{-x}\cos 3x\,dx = -e^{-x}\cos 3x + 3e^{-x}\sin 3x + C$<br>$\int e^{-x}\cos 3x\,dx = \frac{1}{10}e^{-x}(-\cos 3x + 3\sin 3x) + D$<br>$\int_0^{\pi} e^{-x}\cos 3x\,dx = \left[\frac{1}{10}e^{-x}(-\cos 3x + 3\sin 3x)\right]_0^{\pi}$<br>$= \frac{1}{10}e^{-\pi}(-\cos 3\pi + 3\sin 3\pi)$<br>$- \frac{1}{10}e^{0}(-\cos 3(0) + 3\sin 3(0))$<br>$= \frac{1}{10}e^{-\pi} + \frac{1}{10}$ | Remember to include the arbitrary constant once integration is completed |